Final Handsout

MTH 641

FUNCTIONAL ANALYSIS

MODULE # 60 To 113

(FINAL TERM SYLLABUS)

Don't look for someone who can solve your problems,

Instead go and stand in front of the mirror,

Look straight into your eyes,

And you will see the best person who can solve your problems!

Always trust yourself.

(BY ABU SULTAN)

Zain Ahsan

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MODULE NO. 60

LINEAR OPERATORS

➢ *Bounded Linear Operator*

Norms spaces are generalization of distances. By using Norm spaces we are going to discuss Bounded Linear Operator.

Bounded Linear Operator (Definition):

Let X and Y be normed spaces and $T : D(T) \to Y$ a linear operator, where $D(T) \subset X$.

The operator T is said to be bounded if there is a real number c such that for all $x \in D(T)$.

$$\|Tx\| \leq c\|x\|$$

If this condition satisfied then we call T to be a bounded linear operator. Bounded function mean range is bounded but here bounded set is mapping over a bounded set so we call this a bounded linear operator. c is fix.

$$\frac{\|Tx\|}{\|x\|} \leq c \quad , \quad x \in D(T) - \{0\}$$

The smallest possible value of c is supremum of left hand side. Then the value of c is called

$$c = \sup_{\substack{x \in D(T), \\ x \neq 0}} \frac{\|Tx\|}{\|x\|} \qquad \text{as}$$

We call the value as T norm

$$c = \|T\|$$

$$\text{If } D(T) = \{0\}, \quad \|T\| = 0$$

$$c = \|T\| = \sup_{\substack{x \in D(T), \\ x \neq 0}} \frac{\|Tx\|}{\|x\|}$$

$$\|Tx\| \leq \|T\|\|x\|$$

This is the formula that we use for bounded linear operator.

MODULE NO. 61

BOUNDED LINEAR OPERATORS

➢ *Lemma (Norm)*

First we define the norm (equivalent definition) and then prove that the norm defined on T satisfies all four properties of Norm i.e. (N1) to (N4).

Lemma (Statement):

Let T be a bounded linear operator as defined before then an alternate formula for the norm of T is

$$\|T\| = \sup_{\substack{x \in D(T) \\ \|x\|=1}} \|Tx\|$$

The norm defined on T satisfies (N1) to (N4).

Proof: Part (a)

$$\|Tx\| \leq c\|x\|$$

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$$c = \|T\| = \sup_{\substack{x \in D(T), \\ x \neq 0}} \frac{\|Tx\|}{\|x\|} \simeq \sup_{\substack{x \in D(T) \\ \|x\|=1}} \|Tx\|$$

We have to prove

$$\sup_{\substack{x \in D(T), \\ x \neq 0}} \frac{\|Tx\|}{\|x\|} \simeq \sup_{\substack{x \in D(T) \\ \|x\|=1}} \|Tx\|$$

Let $\|x\| = a$; set $y = \frac{x}{a}$, $x \neq 0$,

$$\|y\| = \frac{\|x\|}{a} = 1$$

$$\|T\| = \sup_{\substack{x \in D(T), \\ x \neq 0}} \frac{\|Tx\|}{a}$$

as T is linear so, we take constant a inside the norm

$$\|T\| = \sup_{\substack{x \in D(T), \\ x \neq 0}} \left\| T\left(\frac{1}{a}x\right) \right\| = \sup_{\substack{y \in D(T), \\ \|y\|=1}} \|Ty\| \qquad \text{as } \frac{1}{a} = y$$

Here variable is y which can be any other. Part (a) of lemma is proved.

Part (b):

$$\|T\| = \sup_{\substack{x \in D(T), \\ x \neq 0}} \frac{\|Tx\|}{\|x\|} = \sup_{\substack{x \in D(T), \\ \|x\|=1}} \|Tx\|$$

N1: $\|T\| \geq 0$ is obvious.

N2: $\|T\| > 0 \quad \Rightarrow \quad T=0,$

$$\|T\| = 0 \quad \Rightarrow \quad Tx = 0, \quad \forall \ x \in D(T) \ \Rightarrow \ T = 0$$

N3: $\|\alpha T\| = \sup_{\substack{x \in D(T), \\ \|x\|=1}} \|\alpha Tx\| = \sup_{\|x\|=1} |\alpha| \|Tx\| = |\alpha| \sup_{\|x\|=1} \|Tx\| = |\alpha| \|T\|$ as $\sup_{\|x\|=1} \|Tx\| = \|T\|$

N4: $\|T_1 + T_2\| \leq \|T_1\| + \|T_2\|$

$$\|T_1 + T_2\| = \sup_{\substack{x \in D(T) \\ \|x\|=1}} \|(T_1 + T_2)x\|$$

$$\leq \sup_{\|x\|=1} \|T_1 x + T_2 x\| \leq \sup_{\|x\|=1} \left(\|T_1 x\| + \|T_2 x\| \right)$$

$$= \sup_{\|x\|=1} \|T_1 x\| + \sup_{\|x\|=1} \|T_2 x\| = \|T_1\| + \|T_2\|$$

First we define a $T \times T$ norm and then prove the four properties of norm.

MODULE NO. 62

EXAMPLES BOUNDED LINEAR OPERATORS

- ***Identity Operator***
- ***Zero Operator***
- ***Differentiation Operator***
- ***Integral Operator***

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Identity operator:

$$I : X \to X \qquad \Rightarrow \qquad I_x = x \quad \{x \neq \{0\} \text{ normed space}\}$$

$$\|I\| = \sup_{\substack{x \in D(T), \\ x \neq 0}} \frac{\|Tx\|}{\|x\|} = \sup_{\substack{x \in D(T), \\ x \neq 0}} \frac{\|x\|}{\|x\|} \qquad as \qquad Tx = x$$

$$\|I\| = \sup_{\substack{x \in D(T), \\ x \neq 0}} 1 = 1$$

Zero operator:

The norm space $O : X \to Y$, $O_x = 0$ $\qquad x \in X$

$$\|O\| = \sup_{\substack{x \in D(T), \\ x \neq 0}} \frac{\|Tx\|}{\|x\|} = 0 \quad , \|0\| = 0$$

Differentiation operator:

This is defined on normed space of all polynomial on J=[0, 1]

$$\|x\| = \max\{|x(t)|, \ t \in J\}$$

Value of t varies from 0 to 1 and where the value is maximum, that maximum value is norm of *x*.

applying operator the derivative. Differentiation operator is.

$$Tx(t) = x'(t)$$

Derivation is itself a linear operator.

Now we check that it is bounded or not. $\|Tx(t)\| \leq c\|x(t)\|$. If it is bounded then what is the value of c.

Let $x_n(t) = t^n$ $\quad n \in \mathbb{N}$, what is the norm of $x_n(t)$

$$\|x_n(t)\| = \max\{|x(t)|, \ t \in [0,1]\} = 1$$

Using operator $Tx_n(t) = nt^{n-1}$

define the norm

$$\|Tx_n(t)\| = \max\left|nt^{n-1}\right| = 1$$

$$\|Tx_n(t)\| = \max(|\, nt^{n-1} \,| : t \in [0,1]) = n.1 = n$$

$$\frac{\|Tx_n\|}{x_n} = \frac{n}{1} = c, \quad n \in \mathbb{N}$$

As n had no bound so, there does not exist any c such that $\frac{\|Tx\|}{\|x_n\|} \leq c$ hold.

Now c is fixed number which does not depend upon N but in this case it depends on N, if we take c as n then next value n+1 will not satisfy this equation. It means that there does not exist any c that this condition $\frac{\|Tx\|}{\|x_n\|} \leq c$ holdhence derivative operative is not bounded.

Integral Operator

Defined as $T : C[0,1] \to C[0,1]$,

$$y = Tx \qquad\qquad y(t) = \int_0^1 k(t,\tau)x(\tau)d\tau$$

k is integral of T it is fix for different integral operator,

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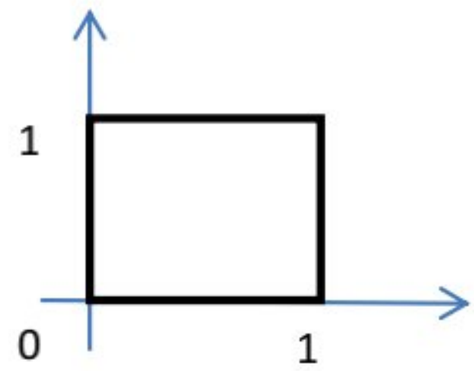
T is linear as integration is linear, also derivation is a linear operator same as integral is linear operator.

K is continuous on $J \times J$. We have two variables t *and* τ , $k(t,\ \tau)$

Whatever the value of k is, it should be in the square

$k(t,\ \tau)$ is bounded. And if it is bounded then

$k(t,\ \tau) \le k_o,\ \ t, \tau \in J \times J\ ,\ \ k_o \in \mathbb{R}$ where $J \times J$ is this square box.



$$\left|x(t)\right| \le \max_{t \in J} \left|x(t)\right| = \left\|x\right\|$$

Now example,

$$\left\|y\right\| = \left\|Tx\right\| = \max_{t \in J} \left| \int_0^1 k(t,\tau)x(\tau)d\tau \right|$$

$$\le \max_{t \in J} \int_0^1 \left|k(t,\tau)\right|\left|x(\tau)\right| d\tau$$

$$\le k_o \left\|x\right\|$$

$\left\|Tx\right\| \le k_o \left\|x\right\|$ it has k *and* k_o is fix so integral operator is a linear operator.

MODULE No. 63

EXAMPLES BOUNDED LINEAR OPERATORS

- ***Matrix***

Identity operator:

$$T : R^n \to R^r$$

$$\begin{bmatrix} a_{11} & . & a_{1n} \\ . & . & . \\ a_{r1} & . & a_{rn} \end{bmatrix} \begin{bmatrix} \xi_1 \\ . \\ \xi_n \end{bmatrix} = \begin{bmatrix} x_1 \\ . \\ x_n \end{bmatrix}$$

$r \times n \qquad n \times 1 \quad r \times 1$

$A \qquad\quad \mathrm{x} \ = \ \mathrm{y}$

The entries are $x = (\xi_j)\ ,\quad y = (\eta_j)$

And the matrix is $A = (\alpha_{ij}),\quad 1 \le \mathrm{i} \le r,\quad 1 \le \mathrm{j} \le n$

$$\eta_j = \sum_{k=1}^{n} \alpha_{jk} \xi k$$

T is linear because the properties of matrices is it bounded?

$$\left\|x\right\| = \left(\sum_{m=1}^{n} \xi_m^2 \right)^{\frac{1}{2}} \qquad ,\quad x \in \mathbb{R}^n$$

and

$$\left\|y\right\| = \left(\sum_{j=1}^{r} \eta_j^2 \right)^{\frac{1}{2}} \qquad ,\quad \mathrm{y} \in \mathbb{R}^n$$

for bounded we have to check norm of T "T(x)".

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Topic # 63

- Examples
- Bounded linear operators:
- Matrix:

$$T: \mathbb{R}^n \longrightarrow \mathbb{R}^r$$

$$\underset{r\times n}{\begin{bmatrix} \alpha_{11} & \cdots & \alpha_{1n} \\ \vdots & & \vdots \\ \alpha_{m1} & \cdots & \alpha_{mn} \end{bmatrix}} \underset{n\times 1}{\begin{bmatrix} \xi_1 \\ \vdots \\ \xi_n \end{bmatrix}} = \underset{r\times 1}{\begin{bmatrix} \eta_1 \\ \vdots \\ \eta_m \end{bmatrix}}$$

$x \mapsto Tx$; $x \mapsto T_x$

$x = (\xi_j)$

$y = (\eta_j)$

$A = (\alpha_{ij})$

$1 \leq i \leq r$

$1 \leq j \leq n$

$x \in \mathbb{R}^n$

$y \in \mathbb{R}^r$

$$Ax = y$$

$$\eta_j = \sum_{k=1}^{n} \alpha_{jk} \xi_k$$

T is linear because the properties of matrix

Is it bounded?

$$\|x\| = \left(\sum_{m=1}^{n} \xi_m^2 \right)^{1/2}$$

$$\|y\| = \left(\sum_{j=1}^{r} \eta_j^2 \right)^{1/2}$$

$$\|y\| = \|Tx\| = \left(\sum_{j=1}^{r} \eta_j^2 \right)^{1/2}$$

$$\Rightarrow \|Tx\|^2 = \sum_{j=1}^{r} \eta_j^2$$

$$\|Tx\|^2 = \sum_{j=1}^{r} \left(\sum_{k=1}^{n} \alpha_{jk} \xi_k \right)^2$$

using Cauchy-Schwarz inequality

$$\leq \sum_{j=1}^{r} \left[\left(\sum_{k=1}^{n} \alpha_{jk}^2 \right)^{1/2} \left(\sum_{m=1}^{n} \xi_m^2 \right)^{1/2} \right]^2$$

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$$= \|x\|^2 \underbrace{\left(\sum_{j=1}^{r}\sum_{k=1}^{n} \alpha_{jk}^2\right)}_{c^2}$$

$$\|Tx\|^2 \leq c^2 \|x\|^2$$

$$\Rightarrow \|Tx\| \leq c\|x\|$$

Topic#64

- Finite Dimension
- Theorem

If a normed space X is finite dimensional, then every linear operator on X is bounded.

- Proof:

X is finite dimension

$\Rightarrow$ every linear operator on X is bounded

Let $\dim X = n \Rightarrow \{e_1, e_2, \ldots, e_n\}$ is a basis for X.

Ld $x \in X$

$$\Rightarrow x = \sum_{j=1}^{n} \xi_j e_j$$

Let T be any linear operator on X.

$$\|Tx\| = \|T\left(\sum_{j=1}^{n} \xi_j e_j\right)\| = \|\sum_{j=1}^{n} \xi_j Te_j\|$$

$$\downarrow \text{linear} \qquad \leq \sum_{j=1}^{n} |\xi_j| \|Te_j\| \leq N_3 \{\|Te_1\|, \|Te_2\|, \ldots, \|Te_n\|\}$$

$$\leq \max \|Te_k\| \sum_{j=1}^{n} |\xi_j| \quad \ldots\ldots (1)$$

$$\|\sum_{j=1}^{n} \xi_j e_j\| \geq c\left(\sum_{j=1}^{n} |\xi_j|\right)^*$$

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$$\sum_{j=1}^{n} |\xi_j| \leq \frac{1}{C} \left\| \sum_{j=1}^{n} \xi_j e_j \right\|$$

$$= \frac{1}{C} \|x\|$$

$$(1) \Rightarrow \|Tx\| \leq \max_{K} \|Te_k\| \frac{1}{C} \|x\|$$

$$\Rightarrow \|Tx\| \leq c' \|x\|$$

$\Rightarrow T$ is bounded.

Topic # 65

- Countinuity and boundedness:
  - Theorem

Let $T: D(T) \to Y$ be a linear operator, where $D(T) \subset X$ and X, Y are normed spaces. Then T is continuous if and only if T is bounded

- Proof:

Let T is continuous at $x_0 \in D(T)$

given $\varepsilon > 0$, $\exists\ \delta$ s.t

$\|T_x - T_{x_0}\| \leq \varepsilon \quad \forall \quad x \in D(T)$ satisfying $\|x - x_0\| < \delta$

Let $y \neq 0 \in D(T)$. we need to show $\|Ty\| \leq c\|y\|$

Let $x - x_0 = \frac{\delta}{\|y\|} y$, since T is linear

$$\|Tx - Tx_0\| = \|T(x - x_0)\| = \left\| T \frac{\delta}{\|y\|} y \right\| = \left\| \frac{\delta}{\|y\|} Ty \right\| = \frac{\delta}{\|y\|} \|Ty\|$$

$$\Rightarrow \frac{\delta}{\|y\|} \|T\| < \varepsilon \quad \because \varepsilon > 0$$

$$\Rightarrow \|Ty\| \leq \frac{\varepsilon}{\delta} \|y\| \quad \because \delta > 0$$

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$$c = \frac{\varepsilon}{\delta}$$

$$\Rightarrow \|Ty\| \leq c\|y\|$$

Conversely suppose that T is bounded.

Let $x_0 \leftarrow D(T)$ (any arbitrary)

To prove continuity, given $\varepsilon > 0$, we need to find $\delta > 0$

$$\|x - x_0\| < \delta$$

$$\|Tx - Tx_0\| < \varepsilon$$

$$\|Tx - Tx_0\| \overset{\text{linear}}{=} \|T(x - x_0)\| \quad \text{bounded } \|T\| = c$$

$$\leq \|T\|\|x - x_0\|$$

$$< \|T\|\delta \qquad \delta = \frac{\varepsilon}{\|T\|}$$

$$= \varepsilon$$

Topic #66

- **Corollary**
- Let T be a bounded linear operator. Then for $x_n, x \in D(T)$
- a: $x_n \to x \Rightarrow Tx_n \to Tx$.
- b: The null space $N(T)$ is closed.

a: $x_n \to x$: T is linear + bounded

we have to show that $Tx_n \to Tx$

$$\|Tx_n - T_x\| = \|T(x_n - x)\| \leq \|T\|\|x_n - x\|$$

$\|x_n - x\| \to 0$

by def $0 < \|Tx_n - Tx\| \leq 0$ $\quad x_n \to x$

$$\Rightarrow \|Tx_n - Tx\| \to 0$$

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$\Rightarrow Tx_n \to Tx$ as $x_n \to x$

(b) $N(T) \subset \overline{N(T)}$ by def.

we need to show

$\overline{N(T)} \subset N(T)$ T (bounded + linear)

$x \in \overline{N(T)}$

$\exists (x_n)$ s.t $x_n \in N(T)$ + $x_n \to x$

by part (a)

$Tx_n \to Tx$

$x_n \in N(T) \forall n \Rightarrow Tx_n = 0 \Rightarrow Tx = 0$

$\Rightarrow x \in N(T)$

$\Rightarrow \overline{N(T)} \subset N(T)$

$\Rightarrow \overline{N(T)} = N(T)$

$\Rightarrow N(T)$ is closed.

Topic # 67

- **Equal Operators :**

Let T_1 and T_2 be two operators, defined to be equal

i.e $T_1 = T_2$ if they have the same domain

$D(T_1) = D(T_2)$

Secondly

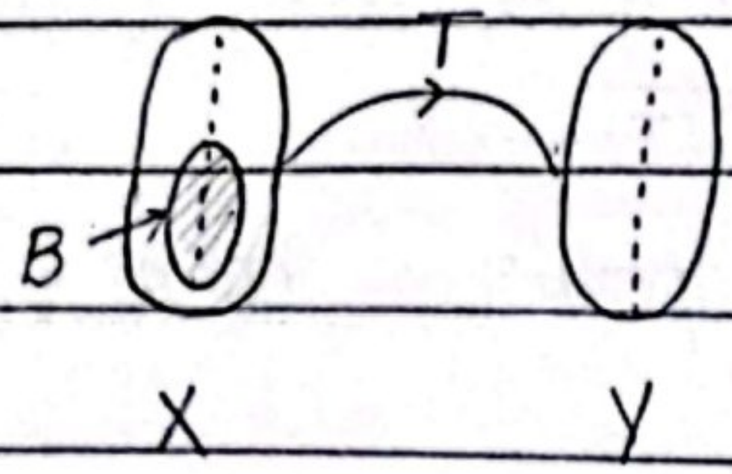
$\forall \; x \in D(T_1) = D(T_2) \Rightarrow T_1 x = T_2 x$

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- Restriction of T:

$T: X \longrightarrow Y$ (any operator)



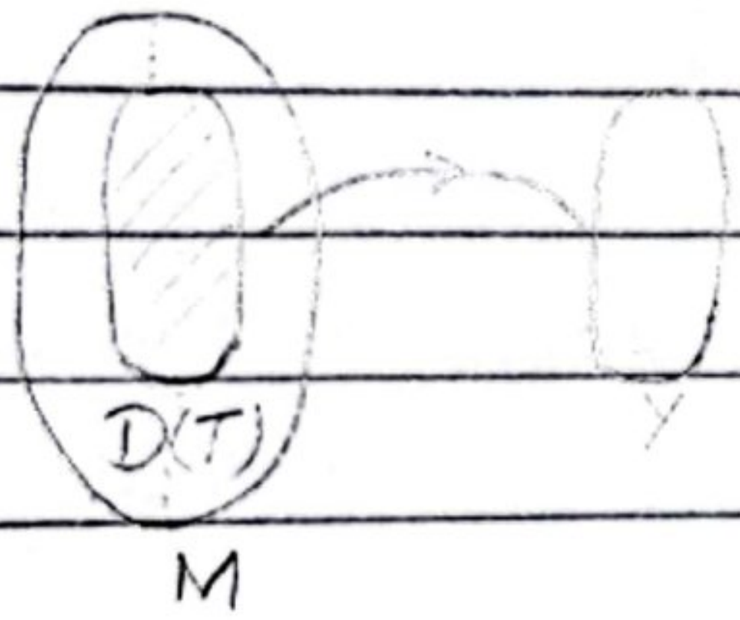
Restriction mean you have a proper subset inside X (Dom) that subset we called B.

$T_{|_B} : B \longrightarrow Y$

$T_{|_B} x = Tx \quad \forall\, x \in B$

- Extension of T:

Extension of T to a set $M \supset D(T)$



$\tilde{T}_{|_{D(T)}} = T$

$\tilde{T} : M \longrightarrow Y$

$\tilde{T}x = Tx \quad \forall\, x \in D(T)$

Topic# 68

- Bounded linear extension:
- Theorem:

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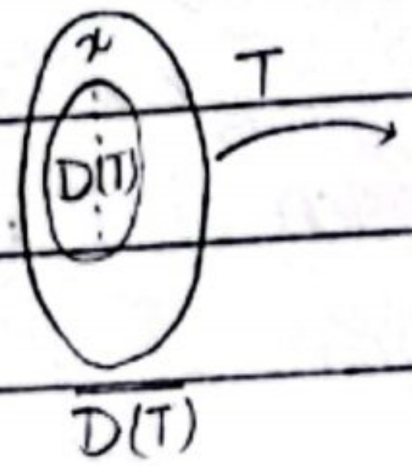
- Let $T: D(T) \to Y$
- be a bounded linear operator, where $D(T)$ lies in a normed space X and Y is a Banach space
- Then T has an extension $\tilde{T}: \overline{D(T)} \to Y$
- Where $\tilde{T}$ is a bounded linear operator of norm $\|\tilde{T}\| = \|T\|$

Proof:

Let $x \in \overline{D(T)}$

$\Rightarrow$ by definition $\exists (x_n)$ in $D(T)$ s.t

$$x_n \to x$$



Since T is linear and bounded, we have

$$\|Tx_n - Tx_m\| \overset{\text{linear}}{=} \|T(x_n - x_m)\| \overset{\text{bounded}}{\leq} \|T\| \|x_n - x_m\|$$

$\Rightarrow (Tx_n)$ is a Cauchy sequence in Y as (x_n) converges

$\Rightarrow (Tx_n)$ converges in Y

Let $Tx_n \to y \in Y$

Define

$$\tilde{T}x = y \qquad x \in \overline{D(T)}$$

Let $x_n \to x$, $z_n \to x$

(v_m) $\quad (x_1, z_1, x_2, z_2, x_3, z_3, \ldots)$

$v_m \to x \Rightarrow$ by previous result (Tv_m) converges to y.

$\Rightarrow$ subsequences (Tx_n), (Tz_n) also converges to same y.

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$\tilde{T}x = Tx \quad \forall\ x \in D(T)$

Also T is bounded, $x_n \in D(T)$

$\Rightarrow \|Tx_n\| \leq \|T\| \|x_n\|$

as $n \to \infty \quad Tx_n \to y = \tilde{T}x$

$\|\check{T}x\| \leq \|T\| \|x\|$

$\Rightarrow \check{T}$ is also bounded

$\|\check{T}\| \leq \|T\| \quad -(a)$

But $\|\check{T}\| \geq \|T\| \quad -(b)$

combining (a) and (b)

$\|\tilde{T}\| = \|T\|$

Topic # 69

• Linear Functionals

A linear functional f is a linear operator with domain in a vector space X and range in the scalar field K of X; thus,

- $f : D(f) \to K$,
- where $K = \mathbb{R}$ if X is real and $K = \mathbb{C}$ if X is complete.

$D(f), R(f), f-, g, h$ bounded linear functional.

A bounded linear function f is a bounded linear operator with range in a scalar field of the normed space X. $\exists\ c$

$$|f(x)| \leq c\|x\|$$

$$\|f\| = \sup_{x \in D(f)} \frac{|f(x)|}{\|x\|} = \sup_{x \in D(f)} |f(x)|$$

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$|f(x)| \leq \|f\| \|x\|$

- **Theorem:**

A linear functional f with domain D(f) in a normed space is continuous (=) f is bounded.

$\longleftarrow \longrightarrow$

Topic # 10

- **Norm**

$\|\cdot\| : X \to \mathbb{R}$

functional but not linear

- **Dot product:**

$f : \mathbb{R}^3 \to \mathbb{R}$

fixed $\underline{a} \in \mathbb{R}^3$ s.t $x \in \mathbb{R}^3$ $\quad \underline{x} = (\xi_1, \xi_2, \xi_3)$

$f(\underline{x}) = \underline{x} \cdot \underline{a}$ $\quad \underline{a} = (a_1, a_2, a_3)$

$= \xi_1 a_1 + \xi_2 a_2 + \xi_3 a_3$

f is linear.

is f is bounded?

$f(\underline{x}) = \underline{x} \cdot \underline{a} \longrightarrow (1)$

$|f(\underline{x})| = |\underline{x} \cdot \underline{a}| \leq \|x\| \|a\|$

$\|x\| = 1$

$\Rightarrow \|f\| \leq \|a\|$

$|f(x)| \leq \|f\| \|x\|$, $x = a$ $\quad |f(a)| = |a \cdot a|$

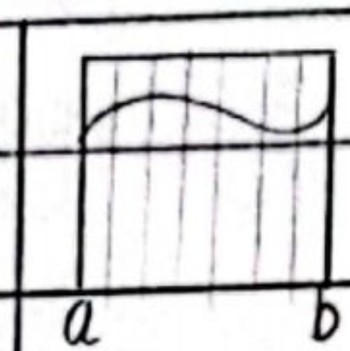
$|f(a)| \leq \|f\| \|a\|$ $\quad = \|a\|^2$

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$$\|f\| \geq \frac{\|a\|^2}{\|a\|} = \|a\| \longrightarrow \|f\| = \|a\|$$

- Definite integral:



$$C[a,b], \quad J=[a,b]$$

$$f(x) = \int_a^b x(t)\,dt$$

$\Rightarrow$ f is linear function.

$$|f(x)| = \left|\int_a^b x(t)\,dt\right| \leq (b-a)\max_{t\in J}|x(t)|$$

$$= (b-a)\|x\|$$

$\|x\| = 1$

$\|f\| \leq b-a \rightarrow (a)$: $\boxed{\|f\| \geq b-a}$

Let $x = x_0 = 1 \Rightarrow \|x_0\| = 1$

$$\|f\| \geq \frac{|f(x_0)|}{\|x_0\|} = |f(x_0)| = \int_a^b 1\,dt = t\Big|_a^b = b-a$$

$\Rightarrow \|f\| \geq b-a \rightarrow (b)$

$\Rightarrow \|f\| = b-a$

Topic # 11

- Examples
- Space $C[a,b]$

$t_0 \in J$ (fixed)

$f_1(x) = x(t_0)$ $\quad x \in C[a,b]$

f_1 is linear, f_1 is bounded $\quad x(t)$

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$$\|Tx\| = \left(\sum_{j=1}^{r} \eta_j^2 \right)^{\frac{1}{2}}$$

$$\|Tx\|^2 = \sum_{j=1}^{r} \eta_j^2$$

$$\|Tx\|^2 = \sum_{j=1}^{r} \left(\sum_{k=1}^{n} \alpha_{jk} \xi_k \right)^2$$

Where $\eta_j = \sum_{k=1}^{n} \alpha_{jk} \xi_k$

Cauchy Schwaz inequality on above $\|Tx\|^2$

$$\leq \sum_{j=1}^{r} \left[\left(\sum_{k=1}^{n} \alpha_{jk}{}^2 \right)^{\frac{1}{2}} \left(\sum_{m=1}^{n} \xi_m^2 \right)^{\frac{1}{2}} \right]^2 = \|x\|^2 \left(\sum_{j=1}^{r} \sum_{k=1}^{n} \alpha_{jk}^2 \right)$$

$$\|Tx\|^2 \leq c^2 \|x\|^2$$

Here is a c which depends upon T.

We can write as

$$\|Tx\| \leq c \|x\|$$

T is already linear and with this value of c we can say matrices is a linear bounded operator.in last four examples three are linear operator but differential was not linear operator.

Module No. 71

Linear Function (Examples):

- Space $C[a \ \ b]$
- Space l^2

Space $C[a \ \ b]$**:**

We have define a linear function on space $C[a \ \ b]$ that we have fixed an element t_o from the set J as $t_o \in J$. Now define a functional operator f(x) which is operating on x which is element from $C[a \ \ b]$. $x \in C[a \ \ b]$

This x is not a variable, it is a function. So f_1 which is defined on $C[a \ \ b]$ linear as it is a linear operator. f_1 is bounded. To find the norm

$$|f_1| = |x(b)| \leq \|x\|$$

$$\|x\| = 1 \quad \Rightarrow \quad \|f_1\| \leq 1 \ldots\ldots\ldots (i)$$

If we take $x_0 = 1$ and substitute in this equation we get

$$|f_1(x_o)| \leq \|f_1\| . \|x\|$$

$$1 \leq \|f_1\| .1 \quad \Rightarrow \quad \|f_1\| \geq 1 \ldots\ldots\ldots (ii$$ From i) and ii)

$$\|f_1\| = 1$$

So the function defined on C is linear, bounded and Norm is 1.

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Space l^2

We choose a fix say $a=(a_j)\in l^2$

$$f(x)=\sum_{j=1}^{\infty}\xi_j a_j \qquad x\in l^2,\ x=(\xi_j)$$

This sequence is linear, converging and bounded.

For boundedness

$$|f(x)|=\left|\sum_{j=1}^{\infty}\xi_j a_j\right|\le\sum_{j=1}^{\infty}|\xi_j a_j|\le\sqrt{\sum_{j=1}^{\infty}|\xi_j|^2}\sqrt{\sum_{j=1}^{\infty}|a_j|^2}=\|x\|.\|a\|$$

It is the same definition of bounded.

M of a complete metric space *X* is itself complete if and only if the set *M* is closed in *X*.

Module No. 72

Linear Function:

- Algebraic Dual Space
- Second Algebraic Dual Space
- Canonical Mapping

Algebraic Dual Space

Set of all linear function defined on a vector space X is itself a vector space and called Algebraic Dual Space and denoted by X^*

Operation on this vector space are

1st Operation Sum

f_1+f_2 $\qquad f_1, f_2$ linear functional

$(f_1+f_2)x=f_1(x)+f_2(x) \quad x\in X$

2nd Operation Scalar Multiplication

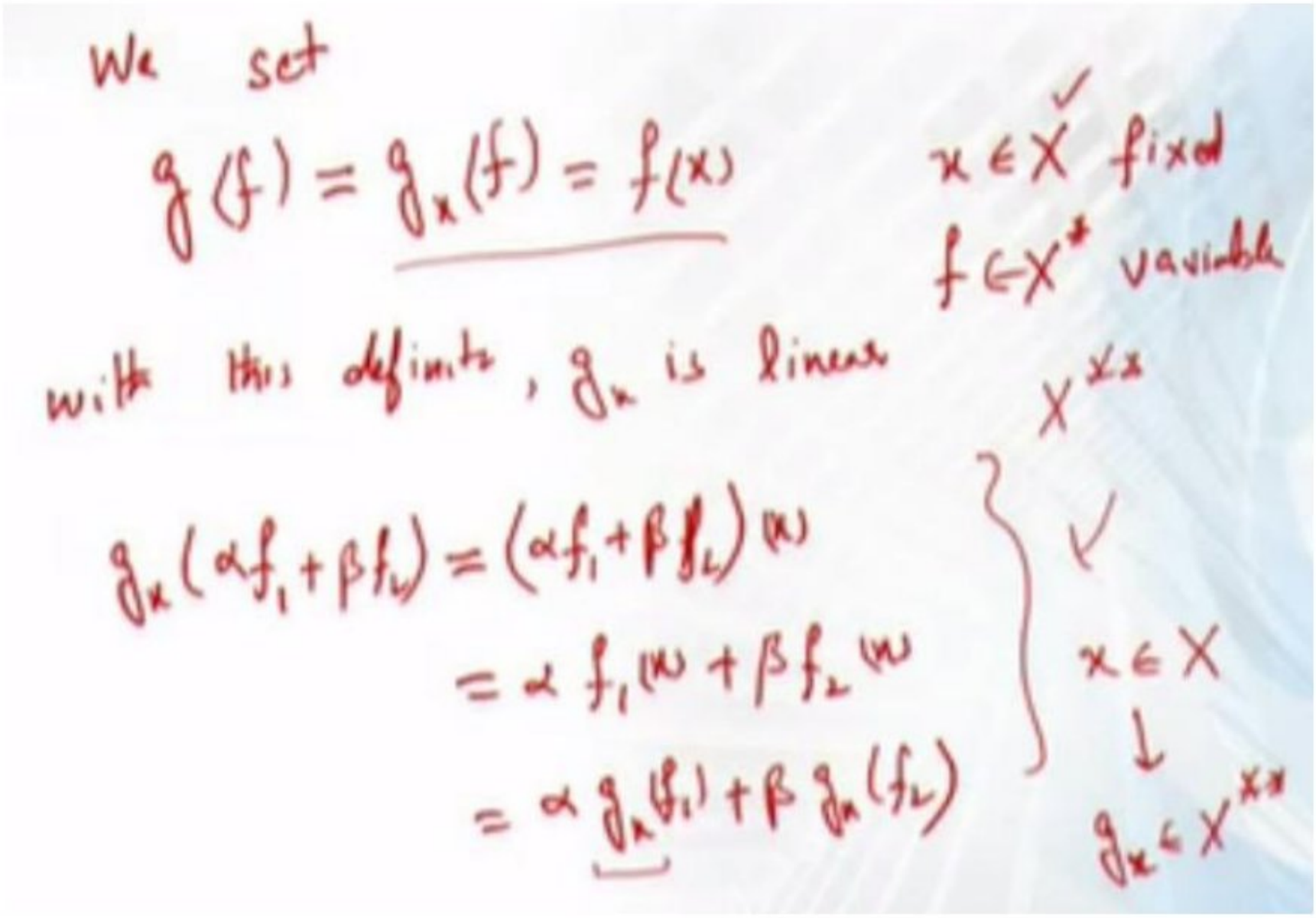
$(af)x=af(x)$

Second Algebraic Dual Space X^{**}

Space	element	Vector at a point
X	$x\in X$	
X^*	g	f (x)
X^{**}	G	g(x)

For each $x, g\in X^{**}$

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Conical Mapping:

$C : X \to X^{**}$ this mapping is called canonical mapping of X into X^{**} defined as $x \mapsto g_x$.

$C(\alpha x + \beta y)(f) = g_{\alpha x+\beta y}(f)$

$= f(\alpha x + \beta y) = \alpha f(x) + \beta f(y) = \alpha g_x(f) + \beta g_{y\partial}(f)$

$= \alpha(Cx)(f) + \beta(Cy)(f)$

So, this is a linear function as well. Canonical mapping is a relation between X and X^{**}.

MODULE NO. 73

LINEAR FUNCTION:

- Algebraically Reflexive
- Second Algebraic Dual Space
- Canonical Mapping

Isomorphism:

It is one-one and onto map.

Algebraically Reflexive:

$T : (X, d) \to (\tilde{X}, \tilde{d})$ bijective

$\tilde{d}(T_x, T_y) = d(x, y)$

$C : X \to X^{**}$ $x \mapsto g_x$.

If C is surjective (on b) bijection. $\Re(C) = X^{**}$

We call X to be algebraically reflexive.

Set of all linear function defined on a vector space X is itself a vector space and called

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MODULE NO. 74

LINEAR OPERATORS AND FUNCTIONAL ON FINITE DIMENSIONAL SPACES:

Finite dimensions mean basis which have finite many elements.

Let X and Y bef.gfinite dimension vector spaces over the same field.

Let $T: X \to Y$ be a linear operator. let $E = \{e_1, ..., e_n\}$ be the basis for X and $B = \{b_1, ..., b_n\}$ be the basis for Y.

$x \in X$, $\quad x = \xi_1 e_1 + \xi_2 e_2 + ... + \xi_n e_n$

$$y = Tx = T\left(\sum_{k=1}^{n} \xi_k e_k\right) = \sum_{k=1}^{n} T\left(\xi_k e_k\right) = \sum_{k=1}^{n} \xi_k T\left(e_k\right)$$

T is uniquely determinal if the image $y_k = Te_k$ of n basis vectors $e_1, ..., e_n$ are prescribed.

$y = Tx \; ; y \in Y \;\; \{b_1, ..., b_n\}$

$y = \eta_1 b_1 + \eta_2 b_2 + ... + \eta_r b_r$

$Te_k \in Y, \quad Te_1 = \tau_{11} b_1 + \tau_{12} b_2 + ... + \tau_{1r} b_r$

$$Te_k = \sum_{j=1}^{r} \tau_{kj} b_j$$

$$y = \sum_{j=1}^{r} \eta_j b_j = \sum_{k=1}^{n} \xi_k Te_k = \sum_{k=1}^{n} \xi_k \sum_{j=1}^{r} \tau_{kj} b_j$$

Comibinig these two summation

$$y = \sum_{j=1}^{r} \left(\sum_{k=1}^{n} \tau_{kj} \xi_k \right) b_j$$

$$\eta_j = \sum_{k=1}^{n} \tau_{kj} \xi_k$$

The image $y = Tx = \sum \eta_j b_j$ of $x = \sum \xi_k Te_k$ can be obtained from

$$\eta_j = \sum_{k=1}^{n} \tau_{kj} \xi_k$$

MODULE NO. 75

OPERATORS ON FINITE DIMENSIONAL SPACES:

Remarks:

As in the case of linear operators on a finite dimensional normed space, every linear functional defined on a finite dimensional normed space is bounded and hence continuous.

Since for linear funcionals range is either $\mathbb{R}$ or $\mathbb{C}$, which are complete. So X^* as the space of all bounded linear functionals defined on X, is also complete and hence is Banach space. This is true even if X is not a Banach space.

"Algebraic Dual Space of X": set of all linear funcionals defined on X.

"Dual or Conjugate Space of X": X^* set of all continuous or bounded linear functionals defined on X.

We take algebraic dual when there is no condition of continuous or bounded linear functions.

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Theorem:

Let X be an n-dimensional vector space and X^* be its dual space. Then

$$\dim X^* = \dim X = n.$$

X^* is collection of linear functions or linear operator while X may be any space.

Proof:

Let $\dim X = n.$

Let basis of X be $B = \{e_1, \ldots\ldots\ldots\ldots, e_2\}$

We define a function.

$$f_j(e_1) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j : \ i, j = 1, \ldots\ldots\ldots, n \end{cases}$$

e.g. j=1 , $f(e_1) = 1, f(e_2) = 0, f(e_3) = 0, \ldots\ldots\ldots\ldots\ldots, f(e_n) = 0$

j=2 , $f(e_1) = 0, f(e_2) = 1, f(e_3) = 0, \ldots\ldots\ldots\ldots\ldots, f(e_n) = 0$

but each n-tuples f_j in this case can be extended as linear functions on X.

MODULE NO. 76

OPERATORS ON FINITE DIMENSIONAL SPACES:

Lemma(Zero Vector):

Let X be a finite deimensional vector space. If $x_0 \in X$ has the property that $f(x_0) = 0$ for all $f \in X^*$ then $x_0 = 0$.

B^* is the basis of X^*

$$\{f_1, f_2, \ldots\ldots, f_n\}$$

$$\Rightarrow f_j(e_i) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$= \delta_{ij}$$

Proof:

For all $x_0 = 0$,

$$x_0 = \sum_{i=1}^{n} x_i e_i \quad ; \quad \mathrm{f} \in \mathrm{X}^* \quad ,$$

$$\mathrm{f}(\mathrm{x}_0) = 0 \quad \Rightarrow \quad \sum_{i=1}^{n} f\left(\sum_{i=1}^{n} x_i e_i\right) = 0$$

$$\Rightarrow \quad \sum_{i=1}^{n} x_i f(e_i) = 0 \quad , \quad \mathrm{j}=1,\ldots\ldots,\mathrm{n}$$

$$\Rightarrow \quad \mathrm{x}_\mathrm{j} = 0 \quad , \quad \forall \ \mathrm{j}=1,\ldots\ldots\ldots,\mathrm{n}$$

$$x_0 = \sum_{i=1}^{n} x_i e_i = 0 \quad \Rightarrow \quad x_0 = \delta$$

MODULE NO. 77

OPERATORS ON FINITE DIMENSIONAL SPACES:

Theorem(Reflexivity):

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A normed space X is said to algebraically reflexive if there is an isometric isomorphism between X and X^{**}.

Ordinarily a normed spacer may not be reflexive.

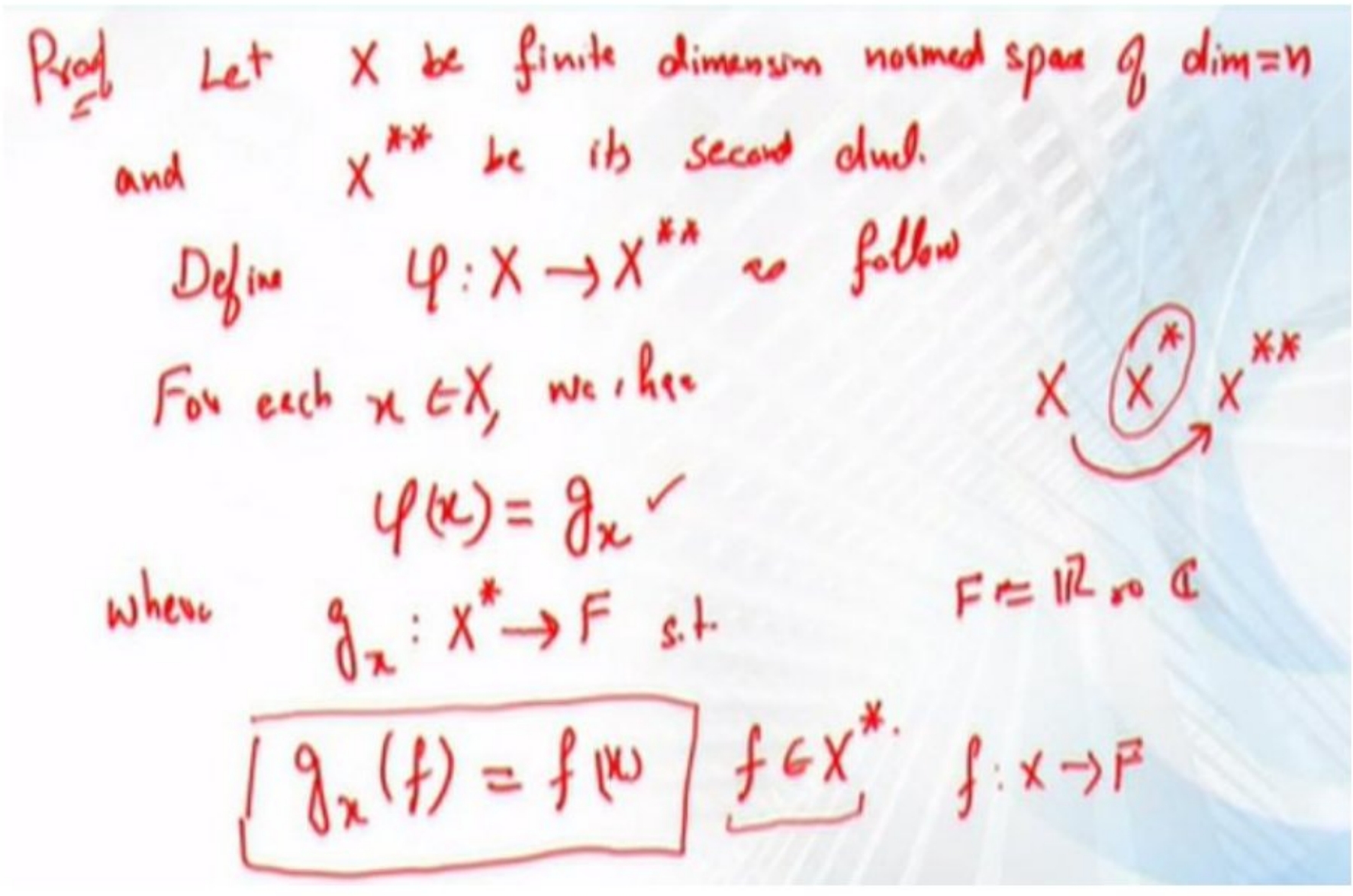
If X is an incomplete normed space even then X^{*} and X^{**} are Banach spaces. So in this case X cannot be a reflexive space.

However there are Banach spaces which are not reflexive.

Theorem:

A finite dimensional vector space is reflexive.

Equivalently, A finite dimensional normed space is isomorphic space is isomorphic to its second dual.



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1) φ is linear

$\varphi : X \to X^{**}$

$\varphi(\alpha x + \beta y) = \alpha\varphi(x) + \beta\varphi(y)$ $\quad x, y \in X$

$\Rightarrow \varphi(\alpha x + \beta y) = g_{\alpha x+\beta y}$

$\varphi(x) = g_x$

$g_x(f) = f(x)$

for $f \in X^*$, $g_{\alpha x+\beta y}(f) = f(\alpha x + \beta y)$

$= \alpha f(x) + \beta f(y)$

$= \alpha g_x(f) + \beta g_y(f)$

$g_{\alpha x+\beta y}(f) = (\alpha g_x + \beta g_y)(f)$

$g_{\alpha x+\beta y} = \alpha g_x + \beta g_y$

$\varphi(\alpha x+\beta y) = \alpha\varphi(x) + \beta\varphi(y)$

2) φ is injective (1-1)

Let $\forall\ x, y \in X$ s.t $\varphi(x) = \varphi(y)$ (we have to show $x = y$)

$\Rightarrow g_x = g_y$

$\Rightarrow g_x - g_y = 0$ ← operator

For $f \in X^*$ $\Rightarrow (g_x - g_y)(f) = 0(f) = 0$ $\quad \forall\ f \in X^*$ (V. Imp)

$\Rightarrow g_x(f) - g_y(f) = 0$

$\Rightarrow f(x) - f(y) = 0 \Rightarrow f(x-y) = 0 \quad \forall\ f \in X^*$

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$\Rightarrow$ by zero lemma

$x - y = 0 \Rightarrow \boxed{x = y}$

φ - linear. $\qquad \varphi : X \to X^{**}$

φ - 1-1 $\Rightarrow X \cong R(\varphi)$

It remains to prove $\boxed{R(\varphi) = X^{**}}$

Now by theorem $\varphi(x) = 0 \Rightarrow x = 0$ ($\Rightarrow \varphi^{-1}$ exist

$\Rightarrow \bar{\varphi} : R(\varphi) \to X$ exist

$\Rightarrow \boxed{\dim(R(\varphi)) = \dim X}$ by the same theorem

if X^* is dual of X, X-f.d

$\dim X = \dim X^*$

applying again $\Rightarrow \dim X^* = \dim X^{**}$

$\Rightarrow \dim X = \dim X^* = \dim X^{**} = \dim(R(\varphi))$

$\dim(X^{**}) = \dim(R(\varphi))$ — (1)

being V.S and (1), $\Rightarrow R(\varphi)$ is not a proper subspace of X^{**}

$R(\varphi) = X^{**}$ $\quad$ φ *is onto*

$X \cong X^{**}$ $\quad$ X *reflexive*

MODULE NO. 78

LINEAR TRANSFORMATION:

Q No.1:

Find the null space of $T : \mathbb{R}^3 \to \mathbb{R}^2$ represented by

$$\begin{bmatrix} 1 & 3 & 2 \\ -2 & 1 & 0 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 3 & 2 \\ -2 & 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} x_1+3x_2+2x_3 \\ -2x_1+x_2 \end{bmatrix}$$

$2\times 3 \quad 3\times 1 \qquad\qquad 2\times 1$

What is meant by null space, it means we have to find those values of $x\in\mathbb{R}^3$ say $x=(x_1,x_2,x_3)$ such that we operate T the answer is zeros as

All those x are element of null space.

$$\begin{bmatrix} x_1+3x_2+2x_3 \\ -2x_1+x_2 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Also we can also say that

$$x_1+3x_2+2x_3=0$$
$$-2x_1+x_2=0$$

We can solve it by using any linear algebra method that will give us solution like echelon form or reduced echelon form and the base of that solution is called basis of null space. Basis mean when apply the element of $\mathbb{R}^3$ the answer should be zero and get a system of linear equation. Find the solution of this system of linear equation. And after finding the solution find the basis that basis are basis of null space.

Example.

Q.NO2

Find the null space of $T:\mathbb{R}^3\to\mathbb{R}^3$ defined by $(\xi_1,\xi_2,\xi_3)\leftrightarrow(\xi_1,\xi_2,-\xi_1-\xi_2)$

1) Basis of $\mathbb{R}(T)$
2) Basis of N(T)
3) Matrix representation.

MODULE NO.79

EXERCISES

DUAL BASIS

Example 1:

a): Find the dual basis of X when basis of X are $B=\{(1,-1,3),(0,1,-1),(0,3,-2)\}$,

Find $B^*=?, X^*=?$ do it yourself

b): let $\{f_1,f_2,f_3\}$ be basis of dual space for X and if X is given by

$e_1=(1,1,1),\ \ e_2=(1,1,-1),\ \ e_3=(1,-1,1)$

Find $f_1(x),f_2(x),f_3(x)$ *when* $x=(0,1,0)$

MODULE NO.80

NORMED SPACES OF OPERATORS

- Examples of Dual Spaces
- $\mathbb{R}^n$

Isometric Isomorphism

A linear operator $\phi:X\to Y.\ X,Y$ normed spaces, is said to be Isometric Isomorphism if

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ϕ isbijective.

ϕ preserve norms.

That is for any

$x \in X$, $\|\phi(x)\| = \|x\|$ is

MODULE NO.81

EXAMPLES SPACES OF OPERATORS

- Examples of Dual Spaces
- l^1

Space l^1

The dual space of l^n is l^∞ means that it is bijective, it is linear and it preserve norm. After defining the map we shall prove these properties one by one.

Proof:

MODULE NO.82

BOUNDED LINEAR OPERATORS

Quiz: Complete norm spaces are called Banach spaces.

Theroem

Let*B(X, Y)* be the set of all bounded linear operators form a normed space *X* to a normed space *Y*.

If ***Y*** is a Banach space, then ***B(X, Y)*** is also a Banach.

Proof:

Let $\{T_n\}$ be an arbitrary Cauchy seq. in *B(X,Y)*.

We will show that $\{T_n\}$converges to an operator *T* in *B(X, Y)*.Since $\{T_n\}$is Cauchy for every

$\varepsilon > 0$ $\exists$ N such that $\|T_n - T_m\| < \varepsilon$ (m,n>N)

For all $x \in X$ and (m,n>N) we have

$$\|T_n(x) - T_m(x)\| = \|(T_n - T_m)(x)\|$$
$$\leq \|T_n - T_m\|\|x\| < \varepsilon\|x\|$$

Thus for a fixed *x* and given $\bar{\varepsilon}$

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Thus for a fixed x and given $\bar{\varepsilon}$ we may choose $\varepsilon = \varepsilon_x$ so that

$$\underbrace{\varepsilon_x \|x\|} < \bar{\varepsilon}$$

$$\Rightarrow \quad \| T_n(x) - T_m(x) \| < \varepsilon \|x\| = \varepsilon_x \|x\| < \bar{\varepsilon}$$

$\Rightarrow \{T_n(x)\}$ is a Cauchy seq. in Y.

Since Y is complete

Let $T_n(x) \xrightarrow{\text{converg}} y$ for some $y \in Y$

We can define a map

$$T : X \to Y$$

$$T(x) = y$$

We will show that T is the required bounded linear operator.

to show ① T is bounded ② $T_n \to T$

T is linear $\quad T(\alpha x + \beta z) \; ; \; x, z \in X$

$T(x) = y, \; T(z) = w$

$$T(\alpha x + \beta z) = \lim_{n \to \infty} T_n(\alpha x + \beta y)$$

$$= \lim_{n \to \infty} T_n(\alpha x) + \lim_{n \to \infty} T_n(\beta y)$$

$$= \alpha \lim_{n \to \infty} T_n(x) + \beta \lim_{n \to \infty} T_n(y)$$

$$= \alpha y + \beta w$$

$$= \alpha T(x) + \beta T(y) \Rightarrow T \text{ is linear.}$$

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1) T is bounded

$$\|T_n(x) - T_m(x)\| < \varepsilon \|x\|$$

$T_n(x) \to y$; $T : X \to Y$ $\quad y = T(x)$

$T_m(x) \to y = T(x)$ $\quad T_n = T(x)$

Thus by continuity of norm we have

$$\|T_n(x) - T(x)\| = \|T_n(x) - \lim_{m\to\infty} T_m(x)\|$$

$$= \lim_{m\to\infty} \|T_n(x) - T_m(x)\| \leq \varepsilon \|x\|$$

$\Rightarrow (T_n - T)$ with $n > N$ is bounded

Also T_n is bounded.

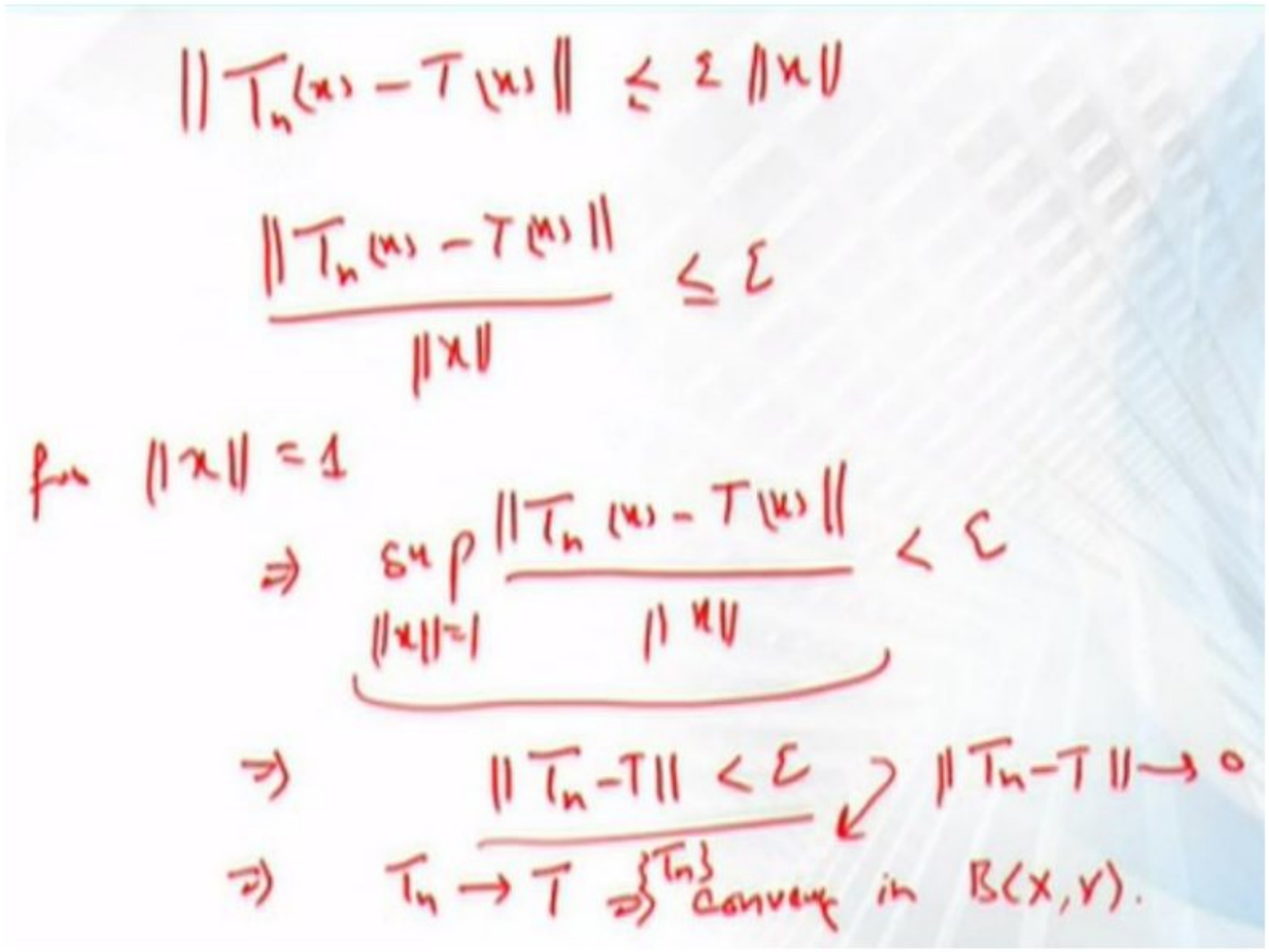
$\Rightarrow T = T_n - (T_n - T)$ ✓ ✓

$\Rightarrow T$ is also bounded

$\Rightarrow T \in B(X, Y)$ $\quad T : X \to Y$

$\{T_n\}$; $T_n \to T$?

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Hence B(X, Y) is complete and Banach space.

Module No.83

Finite Hilbert Spaces

Functional analysis course consist of three major parts parts

1. Metric space is set and we define a space on it that has a certain properties. If it is completer then it is complete space means it should converge within the space
2. Normed Spaces: Norm is a vector space and we define a norm on vector space. Norm is a generalization of distance function.
3. Finite Hilbert Spaces (Inner Product Space)

Hilbert Space

Quiz: Complete inner product space is called a Hilbert Space.

In inner product the generalization is dot product.

Inner product Space

Let V be a vector space over a field F where $F$is $\mathbb{R}$ or $\mathbb{C}$.

An inner product in V is a function $<\bullet,\bullet>: V \times V \to F$ satisfying the following conditions:

Quiz:

Let $x, y, z \in V$; $\alpha \in$ F where α may be real or complex.

i. $< x, x >\geq 0; \quad < x, x >= 0 \quad \Leftrightarrow x = 0$

ii. $< \alpha x, y >= \alpha < x, y >$; but not true for second value as $< x, \alpha y >\neq \alpha < x, y >$

iii. $< x + y, z >=< x, z > + < y, z >$

iv. $< x, y >= \overline{< y, x >}$

$<\bullet,\bullet>: V \times V \to F$ inner product.

Inner Product Space

The pair $(V, <\bullet,\bullet>)$ is called an inner product space.

a): $< ax + by, z > \quad where \quad x, y, z \in V \ , \ a, b \in F$

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Using (iii) property $< ax+by, z >=< ax, z > + < by, z >$

Using (ii) property $a < x, z > +b < y, z >$

$< 0, z >=< 0.x, z >= 0 < x, z >= 0$

b): ***Quiz:***

for all $x, y \in V$, $a \in F$

$< x, ay >= \overline{< ay, x >} = \overline{a < y, x >}$

$= \bar{a}\overline{< y, x >} = \bar{a} < x, y >$

Module No.84

Cauchy Schwarz Inequality

Theorem:

For any two elements x,y is an inner product space V,

$|< x, y >| \le \|x\|.\|y\|$, the define norm is $\|x\| = \sqrt{< x, x >}$, x,y ∈ V

Proof:

If x=y=0 then 0=0

Let at least one of x and y is not equal to zero

Let $|< x+\lambda y, x+\lambda y >| \ge 0$ by definition

$< x, x+\lambda y > + < \lambda y, x+\lambda y >$

$< x, x+\lambda y > + y < y, x+\lambda y >$

Module No.85

Norm on Inner Product space

Theorem:

In an inner product space V, the function $\|\,.\,\| : V \to \mathbb{R}^+$ given by

$\|x\| = \sqrt{< x, y >}$, $x \in V$ defines a norm in V.

Proof:

N1: $\|x\| \ge 0$

For a $x \in V$, $\|x\| = \sqrt{< x, x >} \ge 0$ as $< x, x > \ge 0$

N2:

$\|x\| = 0 \quad \Leftrightarrow \quad \sqrt{< x, x >} = 0 \quad \Leftrightarrow \quad < x, x >= 0 \quad \Leftrightarrow \quad x = 0$

N3: $\|\alpha x\| = |\alpha|\|x\|$

now $\|\alpha x\| = \sqrt{< \alpha x, \alpha x >} \quad \Rightarrow \quad \|\alpha x\|^2 =< \alpha x, \alpha x >$

$\Rightarrow \quad \|\alpha x\|^2 = \alpha\bar{\alpha} < x, x >= |\alpha|^2 \|x\|^2$

N4: $\|x+y\| \le \|x\| + \|y\| \quad \forall$ x, y ∈ V

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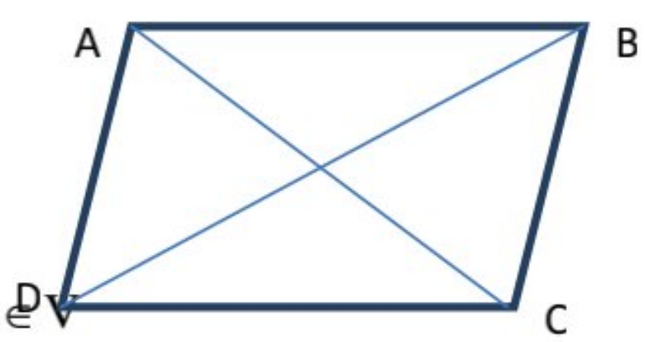
$$\|x+y\|^2 = < x+y, x+y >$$
$$= < x, x+y > + < y, x+y >$$
$$= \overline{< x+y, x >} + \overline{< x+y, y >}$$
$$= \overline{< x, x >} + \overline{< y, x >} + \overline{< x, y >} + \overline{< y, y >}$$
$$= \overline{< x, x >} + \overline{< y, x >} + \overline{< x, y >} + \overline{< y, y >}$$

Now $$= < x, x > + < x, y > + \overline{< x, y >} + < y, y >$$
$$= \|x\|^2 + 2\,\mathrm{Re} < x, y > + \|y\|^2 \qquad \because \mathrm{Re}(z) \le |z|$$
$$\le \|x\|^2 + 2|< x, y >| + \|y\|^2$$
$$\le \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 \qquad \because \ |< x, y >| \le \|x\|\|y\|$$
$$= (\|x\| + \|y\|)^2$$
$$\|x+y\|^2 \le \|x\| + \|y\|$$

MODULE NO.86

PARALLELOGRAM LAW

$$\overline{AC}^2 + \overline{BD}^2 = 2\left(\overline{AB}^2 + \overline{BC}^2\right)$$ **Quiz**



Theorem:

$$\|x+y\|^2 + \|x-y\|^2 = 2\left(\|x\|^2 + \|y\|^2\right)$$ for all x,y ∈ V

Proof:

$$\|x+y\|^2 = < x+y, x+y >$$
$$= < x, x > + < x, y > + \overline{< x, y >} + < y, y >$$
$$= \|x\|^2 + 2\,\mathrm{Re} < x, y > + \|y\|^2 \qquad \text{....(i}$$

Replace y=-y

$$\|x-y\|^2 = < x+y, x+y >$$
$$= < x, x > - < x, y > - \overline{< x, y >} + < y, y >$$
$$= \|x\|^2 - 2\,\mathrm{Re} < x, y > + \|y\|^2 \qquad \text{.....(ii}$$

Adding (i and (ii

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

That we have to prove.

Special Case:

Another result from above equations is

Subtracting (ii from (i

$$\|x+y\|^2 - \|x-y\|^2 = 4\,\mathrm{Re} < x, y >$$

If V is a real inner product space

Re(z)=z or Re<x,y>=<x,y>

$$< x, y > = \frac{1}{4}\left\{\|x+y\|^2 - \|x-y\|^2\right\}$$

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The above form is when V is a real inner product space not complex space.

MODULE NO.87

- POLARIZATION IDENTITY
- APPOLONIUS IDENTITY

Polarization Identity

For any x, y in complex inner product space

$$< x, y >= \frac{1}{4}\left\{\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2\right\}$$

We have to prove this complex inner product space.

Proof:

Let $x, y \in V$

$$\|x+y\|^2 =< x+y, x+y >$$
$$= \|x\|^2 + 2\,\mathrm{Re} < x, y > + \|y\|^2$$
$$= \|x\|^2 + < x, y > + \overline{< x, y >} + \|y\|^2$$
$$= \|x\|^2 + < x, y > + < y, x > + \|y\|^2 \qquad \text{.....(i}$$

If we replace y=-y

$$\|x+y\|^2 = \|x\|^2 + < x, -y > + < -y, x > + \|-y\|^2$$
$$= \|x\|^2 - < x, y > - < y, x > + \|y\|^2 \qquad \text{.....(ii}$$

Replace $y = iy$ in eq(i

$$\|x+iy\|^2 = \|x\|^2 + < x, iy > + < iy, x > + \|iy\|^2$$
$$= \|x\|^2 + \bar{i} < x, y > + i < y, x > + \|y\|^2 \qquad \because \|iy\|^2 =< iy, iy >= i\bar{i} < y, y >= -i^2 < y, y >$$
$$= \|x\|^2 - i < x, y > + i < y, x > + \|y\|^2 \qquad \text{.....(iii}$$

Replace $y = -iy$ in eq(i

$$\|x-iy\|^2 = \|x\|^2 + < x, -iy > + < -iy, x > + \|-iy\|^2$$
$$= \|x\|^2 + i < x, y > - i < y, x > + \|y\|^2 \qquad \text{.....(iv}$$

Subtracting (ii from (i

$$\|x+y\|^2 - \|x-y\|^2 = 4\,\mathrm{Re} < x, y > \qquad \text{.....(v}$$

Subtracting (iv from (iii

$$\|x+iy\|^2 - \|x-iy\|^2 = 2\{i < y, x > - i < x, y >\}$$
$$= -2i\{< x, y > - < y, x >\} = -2i\left\{< x, y > - \overline{< x, y >}\right\}$$
$$= -2i(2i)\,\mathrm{Im} < x, y >= 4\,\mathrm{Im} < x, y > \qquad \text{.....(vi}$$

Now we solve $4\,\mathrm{Re} < x, y > + 4\,\mathrm{Im} < x, y >$

$$\|x+y\|^2 - \|x-y\|^2 + i\|x+y\|^2 - i\|x-y\|^2 = 4\{< x, y >\}$$

Appolonius Identity

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$$\|z-x\|^2+\|z-y\|^2=\frac{1}{2}\|x-y\|^2+2\left\|z-\frac{1}{2}(x+y)\right\|^2 \quad , \quad x,y,z\in V$$

Using parallelogram law

$$\|x'+y'\|^2+\|x'-y'\|^2=2\|x'\|^2+2\|y'\|^2 \quad \text{put} \quad x'=z-x,\ y'=z-y$$

Self-assignment

MODULE NO.88

- **SPACE** $C\left[0,\frac{\pi}{2}\right]$
- **SPACE** l^p

Counter example 1: Space $C\left[0,\frac{\pi}{2}\right]$

Inner product define a norm and under this norm

Every inner product space is a norm space.

Every norm space is not an inner product space. This is not true always.

If a space is inner product then it satisfied the parallelogram law otherwise it is not an inner product space.

We take a norm and built an inner product space and then prove that this inner product space does not satisfy the parallelogram law.

The given set is $C\left[0,\frac{\pi}{2}\right]$ real valued continuous function defined on C[a, b].

The norm of function $f\in C\left[0,\frac{\pi}{2}\right]$, is

$$\|f\|=\sup_{x\in\left[0,\frac{\pi}{2}\right]}|f(x)| \quad ,$$

Let $f,g\in C\left[0,\frac{\pi}{2}\right]$; $f(t)=\sin t,\quad g(t)=\cos t$

We know that sin and cos are continuous functions. Let $C\left[0,\frac{\pi}{2}\right]$ is an inner product space

where the inner product $<\bullet,\bullet>$ define by

$$\|f\|=\sqrt{<f,f>} \quad \Rightarrow \quad <f,f>=\|f\|^2$$

$$\|f\|=\sup_{x\in\left[0,\frac{\pi}{2}\right]}|f(x)|$$

$$\|f+g\|^2+\|f-g\|^2=2\|f\|^2+2\|g\|^2$$

As $f(t)=\sin t,\quad g(t)=\cos t$

$$\|f\|=\sup_{x\in\left[0,\frac{\pi}{2}\right]}|\sin(x)|=1=\|g\|$$

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$$\|f+g\| = \sup_{x\in\left[0,\frac{\pi}{2}\right]} |f(x)+g(x)|$$

$$= \sup_{x\in\left[0,\frac{\pi}{2}\right]} |\sin x+\cos x| = \sqrt{2}$$

$$\|f-g\| = 1$$

Now

$$\|f+g\|^2+\|f-g\|^2 = 2\|f\|^2+2\|g\|^2$$

$$\left(\sqrt{2}\right)^2+(1)^2 = 2\times 1^2+2\times 1^2$$

$$2+1 = 2+2$$

$$3 = 4$$

But $3\neq 4$ so our supposition is wrong. This inner product space does not satisfied parallelogram law. Hence every norm space is not inner product space.

Counter example2: Space l^p

l^p Collection of all bounded sequences,

$P>1, P\neq 2$ if p=2 then it will give inner product space

$$\{x_i\}, \quad \|x\| = \sqrt[p]{\sum_{i=1}^{\infty}|x_i|^p}$$

We will see that $<x,x>=\|x\|^2$ is an inner product space or not. We will check this if it satisfied the parallelogram or not.

Let

$$x=(1,1,0,0,.....) \;;\; y=(1,-1,0,0,.....)$$

$$\|x\| = \sqrt[p]{1^p+1^p+0+0+....} = \sqrt[p]{2} = 2^{\frac{1}{p}}$$

$$\|y\| = \sqrt[p]{1^p+(-1)^p+0+0+....} = \sqrt[p]{2} = 2^{\frac{1}{p}}$$

$$x+y=(2,0,0,0,.....) \quad \Rightarrow \quad \|x+y\| = \sqrt[p]{2^p} = 2^{p\times\frac{1}{p}} = 2$$

$$x-y=(0,2,0,0,.....) \quad \Rightarrow \quad \|x-y\| = \sqrt[p]{2^p} = 2^{p\times\frac{1}{p}} = 2$$

$$\|x+y\|^2+\|x-y\|^2 = 2\|x\|^2+2\|y\|^2$$

$$2^2+2^2 = 2\times 2^{\frac{1}{p}}+2\times 2^{\frac{1}{p}}$$

$$8 = 4\times 2^{\frac{2}{p}} \quad \text{as } p>1,\ p\neq 2$$

The values on both sides are also not equal so this does not satisfied the parallelogram law. Contradict to our supposition. So norm space is not an inner product space.

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MODULE NO.90

➢ THEOREM (CONTINUITY OF INNER PRODUCT)

Theorem:

Let V be any inner product space. For any sequences $\{x_n\}$ and $\{y_n\}$ in V

$$x_n \to x \ , \quad y_n \to y \ \text{ implies} < x_n, y_n > \ \to \ < x, y >$$

Proof:

$$\begin{aligned} &|< x_n, y_n > - < x, y >| \\ &= |< x_n, y_n > - < x_n, y > + < x_n, y > - < x, y >| \\ &= |< x_n, y_n - y > + < x_n - x, y >| \\ &\le |< x_n, y_n - y >| + |< x_n - x, y >| \end{aligned}$$

Now from Cauchy Swarzinequality

$$\begin{aligned} &|x, y| \le \|x\|\|y\| \\ &\le \|x_n\|\|y_n - y\| + \|x_n - x\|\|y\| \end{aligned}$$

Given that $x_n \to x$, $y_n \to y$ so,

$$\|y_n - y\| = \|y - y\| = 0 \quad , \quad \|x_n - x\| = \|x - x\| = 0 \quad \text{as } n \to \infty$$

As $n \to \infty$

$$\begin{aligned} &|< x_n, y_n > - < x, y >| \le 0 \\ &< x_n, y_n > \to < x, y > \quad \text{as } n \to \infty \end{aligned}$$

Theorem:

If $\{x_n\}$ and $\{y_n\}$ are Cauchy sequences in V, then the inner product $< x_n, y_n >$ is a Cauchy sequence in F.

Proof:

$\{x_n\}, \{y_n\}$ are Cauchy sequence

To show $< x_n, y_n >$ is also Cauchy Sequence.

$$\Rightarrow \qquad \|x_n - x_m\| \to 0 \quad ; \quad \|y_n - y_m\| \to 0, \ \ m,n \to \infty$$

$$\begin{aligned} &|< x_n, y_n > - < x_m, y_m >| = |< x_n, y_n > - < x_n, y_m > + < x_n, y_m > - < x_m, y_m >| \\ &= |< x_n, y_n - y_m > + < x_n - x_m, y_m >| \\ &\le |< x_n, y_n - y_m >| + |< x_n - x_m, y_m >| \\ &\le \|x_n\|\|y_n - y_m\| + \|x_n - x_m\|\|y_m\| \end{aligned}$$

$$\Rightarrow \qquad |< x_n, y_n > - < x_m, y_m >| \to 0, \ \text{as} \quad n,m \to \infty$$

$$\Rightarrow \qquad < x_n, y_n > \text{is a Cauchy Sequence}$$

MODULE NO.91

Examples of Inner product spaces

- **SPACE** $\mathbb{R}^n$
- **SPACE** $\mathbb{C}^n$

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- **SPACE** $\mathbb{C}[a,b]$
- **SPACE** l^n
- **SPACE** P_n (Collection of all polynomials of degree n)

Proof:

1. $\mathbb{R}^n$, the elements are of the form

$$x=(x_1,x_2,.....x_n)\ ;\ y=(y_1,y_2,.....y_n)$$

The inner product form is $< x,y >= \sum_{i=1}^{n} x_i y_i$ (Note: check all axiom self-assignment)

The Norm is $\|x\| = \sqrt{< x,x >} = \sqrt{\sum_{i=1}^{n} x_i x_i} = \sqrt{\sum_{i=1}^{n} x_i^2}$

2. $\mathbb{C}^n$

The elements are $z=(z_1,z_2,.....,z_n)\ ;\ z'=(z_1',z_2',.....,z_n')$ if conjugate does not define then it does not satisfied the second or third axiom of inner product space.

The inner product form is $< z,z' >= \sum_{i=1}^{n} z_i \overline{z'}_i$ (Note: check all axiom self-assignment)

3. $\mathbb{C}[a,b]$ be the space of all continuous function defined on [a, b].

$$< f,g >= \int_a^b f(t).\overline{g(t)}dt$$ define an inner product on C[a, b]

(Note: complex function can also be including. In previous example the C[a, b] was not inner product space with define function definition).

$$<\bullet,\bullet>: V\times V \to F$$

We will check all four properties of inner product as

i): $< f,f >=0 \quad \Leftrightarrow \quad f=0$

ii): $< f+g,h >=< f,h > + < g,h >$

iii): $< \alpha f,g >= \alpha < f,g >$

iv): $< g,f >= \overline{< f,g >}$

it define inner product and is define inner product space.

4. l^n is a space of sequences.

$$l^2 : x\{x_i\}$$

The condition or norm is

$$\sum_{i=1}^{\infty} |x_i|^2 < \infty$$

Let defined the inner product of $y=\{y_i\}$ is

$$< x,y >= \sum_{i=1}^{\infty} x_i \overline{y_i}$$

Checd all four axioms as exercise for inner product.

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5. P_n

Let P_n be the collection of all polynomial of degree n(or less than n).

We can write this as $a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a$ e.g $3x^2 - 2x + 1$ of degree two.

Let $u(x), v(x) \in P_n$

The inner product is

$$< u(x), v(x) >= \int_a^b u(x)v(x)dx \quad , \quad x \in [a,b]$$

with this define P_n is an inner product space.

We have not defined conjugate of v(x) as the interval defined is a real valued so its conjugate is also real valued.

MODULE NO.92

Orthogonal Systems

➢ PYTHAGOREAN THEOREM

The dot product of two vectors when they are perpendicular is zero. Similarly in inner product if two vectors are perpendicular then their inner product is zero.

Theorem:

In an inner product space V and x, y in V if $x \perp y$ then

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2$$

Proof:

$$\|x+y\|^2 =< x+y, x+y >$$

$$= < x,x > + < x,y > + < y,x > + < y,y >$$

As x and y are perpendicular so $< x,y >= 0, < y,x >= 0$

$$\|x+y\|^2 =< x,x > + < y,y >= \|x\|^2 + \|y\|^2$$

Generalized form:

$\{x_1, x_2, ...x_n\}$ be nonzero vectors in V inner product space such that

$< x_i, x_j >= 0 \quad , \quad i \neq j$

This system $\{x_1, x_2, ...x_n\}$ is called orthogonal system as all vectors inside it are perpendicular to each other.

The generalized statement is $\|x_1 + x_2 + ... + x_n\|^2 = \|x_1\|^2 + \|x_2\|^2 + ... + \|x_n\|^2$

The idea of proof is

$$\left\|\sum_{i=1}^{n} x_i\right\|^2 = \left\langle \left\|\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i\right\| \right\rangle$$

$$=< x_1 + ... + x_n, x_1 + ... + x_n >$$

$$=< x_1, x_1 + ... + x_n > + ... + < x_n, x_1 + ... + x_n >$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} < x_i, x_j >$$

$< x_i, x_j >= \|x_i\|^2$, if $i \neq j$ $< x_i, x_j >= 0$ and for $i = j$ then $< x_i, x_j >= \|x_i\|^2$

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$$\left\|\sum_{i=1}^{n} x_i\right\|^2 = \sum_{i=1}^{n}\|x_i\|^2$$

MODULE NO.93

Orthogonal Systems

➢ THEOREM (LINEARLY INDEPENDENCE)

Any sequence $\{x_n\}$ of non-zero mutually orthogonal vectors in an inner product space V is linearly independent.

Proof: do it yourself

Let $x = (x_1, x_2, \ldots\ldots x_n)$ be the orthogonal sequence.

Remark:

If $< x, x_1 >= 0$, $\forall$ i=1,2,.....,n $\Rightarrow$ $\left\langle \sum_{i=0}^{n} \alpha_i x_i, x \right\rangle = 0$

$$\left\langle \sum_{i=0}^{n} a_i x_i, x \right\rangle = \left\langle a_1 x_1 + a_2 x_2 + \ldots.. + a_n x_n, x \right\rangle = a_1 \left\langle x_1, x \right\rangle + \ldots\ldots\ldots\ldots + a_n \left\langle x_n, x \right\rangle = 0$$

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inner product

$$\langle e_i, e_j \rangle = \left\langle \frac{x_i}{\|x_i\|}, \frac{x_j}{\|x_j\|} \right\rangle$$

$$= \frac{1}{\|x_i\|\|x_j\|} \langle x_i, x_j \rangle = 0$$

Here are two conditions

i) $i = j$

$$\langle e_i, e_j \rangle = \frac{1}{\|x_i\| \cdot \|x_i\|} \langle x_i, x_i \rangle = \frac{\|x_i\|^2}{\|x_i\|^2} = 1$$

ii) $i \neq j$

$$\langle e_i, e_j \rangle = \frac{1}{\|x_i\|\|x_j\|} \langle x_i, x_j \rangle = \frac{1}{\|x_i\|\|x_j\|} \times 0 = 0$$

$\Rightarrow$ for each orthogonal system $\{x_1, \dots x_n\}$ there exist an orthonormal system give by $\{e_1, \dots e_n\}$ st. $e_i = \frac{x_i}{\|x_i\|}$

we can also say that it is normalige.
normalize mean we chang $\frac{\text{magnitude}}{\text{norm}}$ and this gives value 1
this process is called normalization.

Example #2. (Infinite System)

let $S = \{S_n(t), C_0(t), C_n(t), n = 1, 2, 3, \dots\}$

when $S_n(t) = \frac{1}{\sqrt{\pi}} \sin(nt)$

$C_0(t) = \frac{1}{\sqrt{2\pi}}$

$C_n(t) = \frac{1}{\sqrt{\pi}} \cos(nt)$

} all collection is S

$$\langle x, y \rangle = \int_0^{2\pi} x(t)\, y(t)\, dt.$$

with this inner product, S is an orthonormal system
For orthonormal we prove two properties
inner product is zero for $i \neq j$
inner product is 1 for $i = j$

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$\langle x_2, x_3 \rangle = (2)(2) + (1)(-2) + (-2)(1) = 0.$

As inner product of all vectors x_1, x_2, x_3 is zero, so this is an orthogonal system in R^3

Module # 95

Orthonormal Systems

Def:

A set $\{X_\alpha : \alpha \in \Omega\}$ (collection of vectors) of non zero vectors in an inner product space V is said to be an orthonormal system if

$\langle X_\alpha, X_\beta \rangle = \delta_{\alpha\beta}$

where $\delta_{\alpha\beta}$ is the Kronecker delta

it means $\delta_{\alpha\beta}$ is 0 or 1

if $\alpha\beta$ are same then 1

if $\alpha\beta$ are different then 0

If the inner product of vectors is zero then the system is orthogonal $\langle x_i, x_j \rangle = 0$ when $i \neq j$

If the inner product of vectors is 1 for then it this orthogonal system is orthonormal system

Example #1 (Finite system)

Let $\{x_1, x_2, \ldots, x_n\}$ be any orthogonal system

i.e $\langle x_i, x_j \rangle = 0 \quad \forall \; i \neq j$

Set or take

$e_i = \dfrac{x_i}{\|x_i\|}$ $\|x_i\|$ is norm & real value

$e_j = \dfrac{x_j}{\|x_j\|}$

Now

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inner product

$$\langle e_i, e_j \rangle = \left\langle \frac{x_i}{\|x_i\|}, \frac{x_j}{\|x_j\|} \right\rangle$$

$$= \frac{1}{\|x_i\| \|x_j\|} \langle x_i, x_j \rangle$$

Here are two conditions

i) $i = j$

$$\langle e_i, e_j \rangle = \frac{1}{\|x_i\| \cdot \|x_i\|} \langle x_i, x_i \rangle = \frac{\|x_i\|^2}{\|x_i\|^2} = 1$$

ii) $i \neq j$

$$\langle e_i, e_j \rangle = \frac{1}{\|x_i\| \|x_j\|} \langle x_i, x_j \rangle = \frac{1}{\|x_i\| \|x_j\|} \times 0 = 0$$

$\Rightarrow$ for each orthogonal system $\{x_1, \dots x_n\}$ there exist an orthonormal system give by $\{e_1, \dots e_n\}$ st. $e_i = \frac{x_i}{\|x_i\|}$

we can also say that it is normalize.
normalize mean we change $\frac{\text{magnitude}}{\text{norm}}$ and this gives value 1
This process is called normalization.

Example #2. (Infinite System)

let $S = \{S_n(t), C_0(t), C_n(t), n = 1, 2, 3, \dots\}$

when $S_n(t) = \frac{1}{\sqrt{\pi}} \sin(nt)$

$C_0(t) = \frac{1}{\sqrt{2\pi}}$

$C_n(t) = \frac{1}{\sqrt{\pi}} \cos(nt)$

} all collection is S

$$\langle x, y \rangle = \int_0^{2\pi} x(t)\, y(t)\, dt.$$

with this inner product, S is an orthonormal system
For orthonormal we prove two properties
inner product is zero for $i \neq j$
inner product is 1 for $i = j$

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$\langle S_i, S_j \rangle = 0 \quad i \neq j$

$= 1 \quad i = j$

$\langle C_i, C_j \rangle = 0 \quad i \neq j$

$= 1 \quad i = j$

$\langle S_i, C_j \rangle = 0 \quad \forall \; i \neq j$ or $i = j$

because S & C are different functions

$\langle C_0, C_0 \rangle = 1$

$\langle S_i, C_0 \rangle = \langle C_i, C_0 \rangle = 0 \quad i = 1, 2, \ldots$ C not C_0

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Module #96

Hilbert Space
Theorem (Orthogonalization)

Theorem

Let $x_1, x_2, \ldots, x_n, \ldots$ be any set of linearly independent vector in V, then V contains a set $e_1, e_2, \ldots, e_n, \ldots$ such that $e_1, e_2, \ldots e_m \ldots$ is

i) orthonormal

ii) it generate the same space as generated by $x_1, x_2, \ldots$

Proof:

Lets start with x_1, we define e_1 as

$$e_1 = \frac{x_1}{\|x_1\|} \quad \text{(normalize)}$$

Now multiple of e_1 = multiple of x_1

both have same space

Now $\|e_1\| = \frac{\|x_1\|}{\|x_1\|} = 1$

Let $V_2 = x_2 - \langle x_2, e_1 \rangle e_1$

$$e_2 = \frac{V_2}{\|V_2\|}, \quad \|e_2\| = 1$$

$$\langle e_2, e_1 \rangle = \left\langle \frac{V_2}{\|V_2\|}, e_1 \right\rangle$$

where $\|V_2\|$ is a value and we can write outside of the inner product according to definition

$$= \frac{1}{\|V_2\|} \langle V_2, e_1 \rangle$$

$$\langle e_2, e_1 \rangle = \frac{1}{\|V_2\|} \langle x_2 - \langle x_2, e_1 \rangle e_1, e_1 \rangle$$

$$= \frac{1}{\|V_2\|} \{ \langle x_2, e_1 \rangle - \langle x_2, e_1 \rangle \langle e_1, e_1 \rangle \}$$

$\because \langle e_1, e_1 \rangle = 0$

$$= \frac{1}{\|V_2\|} \{ \langle x_2, e_1 \rangle - \langle x_2, e_1 \rangle 1 \} = 0$$

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$$\{x_1, x_2\} \longrightarrow \{e_1, e_2\}$$

where $e_1 = \frac{x_1}{\|x_1\|}$, $e_2 = \frac{V_2}{\|V_2\|}$, $V_2 = x_2 - \langle x_2, e_1 \rangle e_1$

they generate the same space

$\{x_1, x_2, \dots x_n\} \longrightarrow \{e_1, e_2, \dots, e_n\}$

linear indep orthogonal vectors

$$\langle e_n, e_1 \rangle = \frac{V_n}{\|V_n\|} = \frac{1}{\|V_n\|} \langle V_n, e_1 \rangle$$

$$= \frac{1}{\|V_2\|} \{ \langle x_n, e_1 \rangle - \langle x_n, e_1 \rangle \langle e_1, e_1 \rangle \}$$

$$= \frac{1}{\|V_2\|} \{ \langle x_n, e_1 \rangle - \langle x_n, e_1 \rangle \} = 0$$

Module #97
Hilbert Spaces

Theorem (Riesz and Fischer)

Statement:

Let $\{e_1, \dots, e_n, \dots\}$ be an orthonormal set in a Hilbert space H.

Then for any sequence $\{c_n\}$ of scalar, the following are equivalent.

$\{c_n\} \in l^2$

$\sum_{n=1}^{\infty} c_n e_n$ converges in H

there is an element $x \in H$ such that $\langle x, e_n \rangle = c_n$, $n = 1, 2, \dots$

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i) ⇒ (ii) if i) is true then ii) is true.

$\{c_n\} \in l^2 \Rightarrow \sum_{k=1}^{\infty} |c_n|^2 < \infty$ (given)

sequence of scalar — we have to prove $\sum_{n=1}^{\infty} c_n e_n$ converges ← and we prove every series of this

set $S_n = \sum_{k=1}^{n} c_n e_n$

We will show that $\{S_n\}$ is a Cauchy seq.
We consider that H is a complet space and every seq. in complete space is converging in it and if we prove that it is a Cauchy seq. then automatically it proved that it is converging. So we have to just prove that $\{S_n\}$ is Cauchy seq. in H

Let $\|S_m - S_n\|^2 = \langle S_m - S_n, S_m - S_n \rangle$ as $\|x\| = \sqrt{\langle x, x \rangle}$

$= \left\langle \sum_{k=1}^{m} c_n e_n - \sum_{k=1}^{n} c_n e_n , \sum_{n=1}^{m} c_n e_n - \sum_{n=1}^{n} c_n e_n \right\rangle$

Expanding from m, n term will cancel out)

$= \left\langle \sum_{n=n+1}^{m} c_n e_n , \sum_{n=n+1}^{m} c_n e_n \right\rangle$

For orthonormal system $e_i \cdot e_j = \delta_{ij} = 1 \quad i = j$

$\delta_{ij} = 0 \quad i \neq j$

Now $\|S_m - S_n\|^2 = \sum_{n=n+1}^{m} e_n \bar{c}_n = \sum_{n=n+1}^{m} |c_n|^2$

but given $\sum_{n=1}^{\infty} |c_n|^2$ converges

$\Rightarrow \sum_{n=n+1}^{m} |c_n|^2 \to 0$ as $m, n \to \infty$

$\Rightarrow \{S_n\}$ is Cauchy seq proved

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ii) $\Rightarrow$ iii)

$\sum_{n=1}^{\infty} c_n e_n$ converges in i)

let $x \in H \Rightarrow x = \sum_{n=1}^{\infty} c_n e_n$

$\Rightarrow \langle x, e_i \rangle = \left\langle \sum_{n=1}^{\infty} c_n e_n, e_i \right\rangle$

$= \sum_{n=1}^{\infty} c_n \langle e_n, e_i \rangle$

Apply orthonormal property $i=n, \langle e_i, e_i \rangle = 1$

$\Rightarrow \langle x, e_i \rangle = c_i$

iii) $\Rightarrow$ i)

given $\left.\begin{matrix} \\ \end{matrix}\right\}$ $x \in H$ s.t. $\langle x, e_i \rangle = c_i$, $i = 1, 2, \cdots$

prove $\{c_n\} \in l^2$

$\sum_{n=1}^{\infty} |c_i|^2 = \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2$

by Bessels inequility

$\leq \|x\|^2$

$< \infty$

which mens that it is bounded

$\Rightarrow \sum_{n=1}^{\infty} |c_n|^2 < \infty$

$\Rightarrow \{c_n\} \in l^2$ proved

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Module # 98

The Gram-Schmidt Orthogonalization process

Let V be a Vector Space with an inner space. Let

$x_1, \ldots, x_n$ be a basis for V

Orthonormal

Orthonormal { Ortho means we find orthogonal basis
normal means its norm is 1

So, [Any basis] → [Orthogonal] → [normalize]

i) Ortho. $V_1 = x_1$

$$V_2 = x_2 - \frac{\langle x_2, V_1 \rangle}{\langle V_1, V_1 \rangle} V_1$$

$$V_3 = x_3 - \frac{\langle x_3, V_1 \rangle}{\langle V_1, V_1 \rangle} V_1 - \frac{\langle x_3, V_2 \rangle}{\langle V_2, V_2 \rangle} V_2$$

$$\vdots$$

$$V_n = x_n - \frac{\langle x_n, V_1 \rangle}{\langle V_1, V_1 \rangle} V_1 - \cdots - \frac{\langle x_n, V_{n-1} \rangle}{\langle V_{n-1}, V_{n-1} \rangle} V_{n-1}$$

where $V_1, V_2, \ldots, V_n$ is an orthogonal basis for V.

ii) For Normalize

Let $\{V_1, \ldots, V_n\}$ is an orthogonal

We define

$$w_1 = \frac{V_1}{\|V_1\|}, \quad w_2 = \frac{V_2}{\|V_2\|}, \ldots, \quad w_n = \frac{V_n}{\|V_n\|}$$

$\{w_1, w_2, \ldots, w_n\}$ is an orthonormal basis for V.

so, above are also orthogonal basis from given basis

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Theorem:

Any finite dimensional vector space with an inner product has an orthonormal basis.

Exampl $\mathbb{R}^3$

$x_1 = (1,2,2)$: $x_2 = (-1,0,2)$

V_0 is spanned by x_1, x_2

i) Find orthonormal basis for V_0

ii) Extend it to an orthonormal basis for $\mathbb{R}^3$

we take standard basis for $x_3 = (0,0,1)$

Now for basis, vectors should be linearly independent.

Check linearly Independent.

From linear Algebra

if the determinant of these vector is not zero

e.g. $$\begin{vmatrix} 1 & 2 & 2 \\ -1 & 0 & 2 \\ 0 & 0 & 1 \end{vmatrix} \neq 0$$

$$2 \neq 0$$

Now **Applying process**

$x_1 = (1,2,2)$, $x_2 = (-1,0,2)$, $x_3 = (0,0,1)$

$V_1 = x_1 = (1,2,2)$

$$V_2 = x_2 - \frac{\langle x_2, V_1 \rangle}{\langle V_1, V_1 \rangle} V_1 = (-1,0,2) - \frac{3}{9}(1,2,2)$$

$$= (-1,0,2) - \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

$$= \left(-\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)$$

where $\langle x_2, V_1 \rangle = (-1)(1) + (0)(2) + (2)(2) = -1 + 0 + 4 = 3$

$\langle V_1, V_1 \rangle = (1)(1) + (2)(2) + (2)(2) = 1 + 4 + 4 = 9$

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$$V_3 = x_3 - \frac{\langle x_3, V_1\rangle}{\langle V_1, V_1\rangle} V_1 - \frac{\langle x_3, V_2\rangle}{\langle V_2, V_2\rangle} V_2$$

$$= (0,0,1) - \frac{2}{9}(1,2,2) - \frac{4/3}{4}\left(-\frac{4}{3}, \frac{-2}{3}, \frac{4}{3}\right)$$

$$= \left(\frac{2}{9}, \frac{-2}{9}, \frac{1}{9}\right)$$

This gives the orthogonal system in $\mathbb{R}^3$

For Orthonormal.

For orthonormal we first have to calculate the norm of $V_1, V_2, --, V_n$ and norm is calculated as

$$\|V\| = \sqrt{\langle V, V\rangle}$$,(norm = $\sqrt{\text{inner product}}$

Now $\|V_1\| = \sqrt{\langle V_1, V_1\rangle}$

$$= \sqrt{\langle (1,2,2), (1,2,2)\rangle}$$

$$= \sqrt{(1)(1)+(2)(2)+(2)(2)} = \sqrt{9} = 3$$

$$\|V_2\| = \sqrt{\langle V_2, V_2\rangle}$$

$$= \sqrt{\left\langle \left(-\frac{4}{3}, \frac{-2}{3}, \frac{4}{3}\right), \left(-\frac{4}{3}, \frac{-2}{3}, \frac{4}{3}\right)\right\rangle}$$

$$= \sqrt{\frac{16}{9} + \frac{4}{9} + \frac{16}{9}} = \sqrt{\frac{36}{9}} = 2$$

Similarly $\|V_3\| = \sqrt{\langle V_3, V_3\rangle} = \sqrt{\frac{1}{9}} = \frac{1}{3}$

So, the orthonormal system

$$w_1 = \frac{V_1}{\|V_1\|} = \frac{(1,2,2)}{3}$$

Similarly $w_2 = \frac{V_2}{\|V_2\|} = \frac{1}{2}\left(-\frac{4}{3}, \frac{-2}{3}, \frac{4}{3}\right)$

$$w_3 = \frac{V_3}{\|V_3\|} = \frac{1}{1/3}\left(\frac{2}{9}, \frac{-2}{9}, \frac{1}{9}\right)$$

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Module # 99

Example

Distance of a point and a subspace

Problem

Find the distance from the point $y = (0, 0, 0, 1)$ to the subspace $V \subset R^4$ spanned by vectors

$x_1 = (1, -1, 1, -1)$, $x_2 = (1, 1, 3, -1)$

$x_3 = (-3, 7, 1, 3)$ three dimension

Mechanism:

We apply Gram - Schmidt process to the vectors x_1, x_2, x_3, y which give us orthogonal system, which means all vectors are orthogonal to each other, so out put is

$x_1, x_2, x_3, y \xrightarrow[\text{output}]{} V_1, V_2, V_3, V_4$

Required distance is $|V_4|$

For V_1 $\quad V_1 = x_1 = (1, -1, 1, -1)$

For V_2 $\quad V_2 = x_2 - \frac{\langle x_1, V_1 \rangle}{\langle V_1, V_1 \rangle} V_1 = (1, 1, 3, -1) - \frac{(1-1+3+1)}{(1+1+1+1)} (1, -1, 1, -1)$

$= (1, 1, 3, -1) - \frac{4}{4}(1, -1, 1, -1) = (0, 2, 2, 0)$

$= ((1-1), 1-(-1), (3-1), -1-(-1))$

For V_3 $\quad V_3 = x_3 - \frac{\langle x_3, V_1 \rangle}{\langle V_1, V_1 \rangle} V_1 - \frac{\langle x_3, V_2 \rangle}{\langle V_2, V_2 \rangle} V_2$

$= (-3, 7, 1, 3) - \frac{(-3-7+1+3)}{(1+1+1+1)} (1, -1, 1, -1) - \frac{(0+14+2+0)}{(0+4+4+0)} (0, 2, 2, 0)$

$= (-3, 7, 1, 3) + (3, -3, 3, -3) - (0, 4, 4, 0)$

$= ((-3+3+0), (7-3-4), (1+3-4), (3-3-0))$

$= (0, 0, 0, 0)$

We are looking for non-zero vector but this is a zero vector

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It shows that x_3 is a linear combination of x_1 and x_2 which means that this is not a space, this is a plan. It is 2-dimension.

Note:

We can use Gram-Schmidt process to check linear independence

Now we have plan so, the system can be revised as

$$x_1, x_2, y \Longrightarrow V_1, V_2, \tilde{V}_3 \quad (\sim \text{Telda})$$

Required distance is $|\tilde{V}_3|$

Now V_1 and V_2 already be found just need to find $\tilde{V}_3$

$$\tilde{V}_3 = y - \frac{\langle y, V_1\rangle}{\langle V_1, V_1\rangle} V_1 - \frac{\langle y, V_2\rangle}{\langle V_2, V_2\rangle} V_2$$

$$= (0,0,0,1) - \frac{(0+0+0-1)}{1+1+1+1}(1,-1,1,-1) - \frac{(0+0+0+0)}{(0+4+4+0)}(0,2,2,0)$$

$$= (0,0,0,1) + \left(\frac{1}{4}, -\frac{1}{4}, +\frac{1}{4}, -\frac{1}{4}\right) - (0,0,0,0)$$

$$= \left(0+\frac{1}{4}, 0-\frac{1}{4}, 0+\frac{1}{4}, 1-\frac{1}{4}\right) = \left(\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, +\frac{3}{4}\right)$$

$$|\tilde{V}_3| = \left|\frac{1}{4}, -\frac{1}{4}, \frac{1}{4}, +\frac{3}{4}\right|$$

$$= \frac{1}{4}|(1,-1,1,+3)| = \frac{1}{4}\sqrt{(1+1+1+9)}$$

$$= \frac{1}{4}\sqrt{12} = \frac{\sqrt{12}}{4} = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$$

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Module #100

Annihilators

orthogon

These are basically subset of Hilbert space with respect to inner product to all other elements of Hilbert space

Def:

Let H be a Hilbert space and $A \subset H$, For any $x \in H$

we say x is orthogonal to A ($x \perp A$)

orthogonal sign

if and only if $\langle x, y \rangle = 0 \quad \forall \quad y \in A$

inner product is generalization of dot product and dot product is when they are perpendicular to each other

The set of all vectors orthogonal to A is called annihilator of A.

we denot it by $A^{\perp}$ or $A^{\perp} = \{x \in H : x \perp A\}$

$A^{\perp}$ inner product A is zero

$\langle A^{\perp}, A \rangle = 0$

Mathematically:

$$A^{\perp} = \{x \in H : x \perp A\}$$

Now for $A^{\perp\perp}$

$A^{\perp\perp}$ is annihilator of $A^{\perp}$

by definition $\{0\}^{\perp} = H$

For whole Hilbert space

$$\{H\}^{\perp} = \{0\}$$

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Module # 101

Theorem

Let A and B be subsets of a Hilbert space H. Then

i) $A \subseteq A^{\perp\perp}$ A contained in $A^{\perp\perp}$ ~~not~~ $A^{\perp\perp} \neq A$

ii) $A \subseteq B \Rightarrow B^{\perp} \subseteq A^{\perp}$

iii) $(A \cup B)^{\perp} = A^{\perp} \cap B^{\perp}$

iv) $A^{\perp} \cup B^{\perp} \subseteq (A \cap B)^{\perp}$

v) $A^{\perp} = A^{\perp\perp\perp}$

vi) $A \cap A^{\perp} \subseteq \{0\}$

vii) $A^{\perp}$ is a closed subspace of H

Proof: i) $A \subseteq A^{\perp\perp}$

Let $x \in A$, $\Rightarrow \langle x, y \rangle = 0 \quad \forall \; y \in A^{\perp}$

$\Rightarrow y \in A^{\perp}$ and $\langle x, y \rangle = 0$

$\Rightarrow$ ~~xx~~ $x \in (A^{\perp})^{\perp} = A^{\perp\perp}$

so if we take any element from A say x it should be in $A^{\perp\perp}$.

ii) $A \subseteq B \Rightarrow B^{\perp} \subseteq A^{\perp}$

Proof:

Now $x \in B^{\perp}$

$\Rightarrow \langle x, y \rangle = 0 \quad \forall \quad y \in B$

$A \subseteq B \Rightarrow \langle x, y \rangle = 0 \quad \forall \; y \in A$

$\Rightarrow x \in A^{\perp}$

so we take $x \in B^{\perp}$ and get $x \in A^{\perp}$

this shows that $B^{\perp} \subseteq A^{\perp}$ $B^{\perp}$ contained in $A^{\perp}$

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iii) $(A \cup B)^{\perp} = A^{\perp} \cap B^{\perp}$

Since $A \subseteq A \cup B$ and $B \subseteq A \cup B$

Applying $A \subseteq B \Rightarrow B^{\perp} \subseteq A^{\perp}$

So, $A \subseteq A \cup B \Rightarrow (A \cup B)^{\perp} \subseteq A^{\perp}$

$B \subseteq A \cup B \Rightarrow (A \cup B)^{\perp} \subseteq B^{\perp}$

this shows that if $(A \cup B)^{\perp}$ is in both $A^{\perp}, B^{\perp}$ so $(A \cup B)^{\perp}$ must be in intersection of $A^{\perp}, B^{\perp}$

$$(A \cup B)^{\perp} \subseteq A^{\perp} \cap B^{\perp} \quad \text{---- (i)}$$

Now we prove that $A^{\perp} \cap B^{\perp} \subseteq (A \cup B)^{\perp}$

let $x \in A^{\perp} \cap B^{\perp} \Rightarrow x \in A^{\perp}, x \in B^{\perp}$

It means

$\langle x, y \rangle = 0 \quad \forall \ y \in A,$

and $\langle x, w \rangle = 0 \quad \forall \ w \in B$

$\Rightarrow \langle x, v \rangle = 0 \quad \forall \ v \in A \cup B \Rightarrow x \in (A \cup B)^{\perp}$

Now we take

$x \in A^{\perp} \cap B^{\perp}$ and get $x \in (A \cup B)^{\perp}$

$\Rightarrow A^{\perp} \cap B^{\perp} \subseteq (A \cup B)^{\perp}$ ---- (ii)

from i) and ii)

$$(A \cup B)^{\perp} = A^{\perp} \cap B^{\perp}$$

iv) $A^{\perp} \cup B^{\perp} \subseteq (A \cap B)^{\perp}$

let A, we know

$A \cap B \subseteq A \Rightarrow A^{\perp} \subseteq (A \cap B)^{\perp}$ (prop ii)

$A \cap B \subseteq B \Rightarrow B^{\perp} \subseteq (A \cap B)^{\perp}$ (prop ii)

so, $A^{\perp} \cup B^{\perp} \subseteq (A \cap B)^{\perp}$

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v)

From prop i) $A \subseteq A^{\perp\perp}$

From prop ii) $(A^{\perp\perp})^{\perp} \subseteq A^{\perp}$

$\Rightarrow$ $A^{\perp\perp\perp} \subseteq A^{\perp}$ — i

Again by i)

$A^{\perp} \subseteq (A^{\perp\perp})^{\perp}$ — ii

as this is true for $A, A^{\perp}, A^{\perp\perp}$ using again

from prop ~~ii~~ i) we take A^* as $A^{\perp}$ and $(A^{\perp\perp})^{\perp}$ as $A^{\perp\perp}$

from i) and $A^{\perp} = A^{\perp\perp\perp}$

vi) $A \cap A^{\perp} \subseteq \{0\}$

Case a) if $A \cap A^{\perp} = \phi$ (empty)

$\Rightarrow$ $A \cap A^{\perp} \subseteq \{0\}$

Case b)

Let $x \in A \cap A^{\perp}$

$x \in A$; $x \in A^{\perp}$

$x \perp x$

$\langle x, x \rangle = \|x\|^2 = \quad \forall \; x \in A$

which is contradiction

vii) $A^{\perp}$ is a closed subspace of H.

First prove that $A^{\perp}$ is a subspace

Second prove that $A^{\perp}$ is closed

closed means any limit point of $A^{\perp}$ is in $A^{\perp}$

For subspace

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Module # 102

Direct Decomposition

Theorem:

Result 1

Let A be a non-empty complete convex subset of an inner product space V, and $x \in V \setminus A$

Then there is a unique $y \in A$ which is closed to x

Result 2

Let A be a complete subspace of an inner product space V.

Then there is a non-zero vector z in V such that

$$z \perp A \qquad (z \text{ is perpendicular } A)$$

Theorem:

Let A be a proper complete subspace of an inner product space V.

Then

$$V = A \oplus A^{\perp}$$

Proof:

To show each element of $x \in V$ is of the form $x = y + z$

$$y \in A, \; z \in A^{\perp}$$

this expression is unique

there exist $z \perp A \Rightarrow z \in A^{\perp}$

$z = x - y$, $x = y + z$ using result 1 & 2

To prove it a unique

Let $x = y_1 + z_1$, $y_1 \in A$, $z_1 \in A^{\perp}$

$x = y_2 + z_2$ $y_2 \in A$, $z_2 \in A^{\perp}$

$\Rightarrow \underbrace{y_1 - y_2}_{\text{contained in } A} = \underbrace{z_2 - z_1}_{\text{contained in } A^{\perp}} \in A \cap A^{\perp} = \{0\}$

$\Rightarrow y_1 - y_2 = 0 \Rightarrow y_1 = y_2$

$z_1 - z_2 = 0 \Rightarrow z_1 = z_2$

$V = A \oplus A^{\perp}$ expression is unique

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Module # 103

Annihilators

Theorem:

Let A and B be closed subspace of a Hilbert space H such that $A \perp B$. Then $A+B$ is a closed subspace of H.

Now if A and B is a subspace then $A+B$ is also a subspace. We just need to prove that $A+B$ is a closed subspace

Two method, limit point or Cauchy seq. converge

Proof:

we need to show only $A+B$ is closed.

Let z be a limit point of $A+B$.

$\Rightarrow$ there exist a sequence $\{z_n\}$ in $A+B$, such that

$$z = \lim_{n\to\infty} z_n$$

where $z_n = x_n + y_n$, $x_n \in A$, $y_n \in B$

we need to show that $\{x_n\}$ and $\{y_n\}$ are Cauchy seq in A and B

{let say $\{x_n\} \to x$ $\{y_n\} \to y$ not include}

$z = x + y = A + B$)

by pythagorean theorem

$$\|z_m - z_n\|^2 = \|x_m + y_m - x_n - y_n\|^2$$

$$= \|x_m - x_n + y_m - y_n\|^2$$

$$= \|x_m - x_n\|^2 + \|y_m - y_n\|^2$$

Since $\{z_n\}$ is a Cauchy seq

$\Rightarrow$ $\{x_n\}$ and $\{y_n\}$ are also Cauchy seq

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Module # 104

Hilbert-adjoint Operator T^*

This Hilbert-adjoint operator is used to solve problem in matrices, differential equations and integral equations.

Hilbert-adjoint Operator T^*

Let $T: H_1 \longrightarrow H_2$ be a bounded linear operator, where H_1 and H_2 are Hilbert spaces. Then the Hilbert adjoint operator T^* of T is the operator

$$T^*: H_2 \longrightarrow H_1 \qquad (T: H_1 \longrightarrow H_2)$$

such that for all $x \in H_1$ and $y \in H_2$

$$\langle Tx, y \rangle = \langle x, T^* y \rangle$$

if this condition is hold then T^* is Hilbert adj op

Example:

Let A be a $m \times n$ real matrix

$x \longmapsto Ay$ be a linear map.

$(\mathbb{R}^n \longrightarrow \mathbb{R}^m)$

$\forall\ x \in \mathbb{R}^n,\ y \in \mathbb{R}^m$

$Ax \cdot y = (Ax)^T y \qquad x \cdot y = x^T y$

$= x^T A^T y$

$= x \cdot (A^T y)$ A is a operator

Hilbert-adjoint operator is A^T

Theorem:

The Hilbert-adjoint operator T^* of T in above definition exists, is unique and is a bounded linear operator with norm

$$\|T^*\| = \|T\|$$

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Sesquilinear form

Let X and Y be vector space over the same field $X(\mathbb{R}, \mathbb{C})$.

Then the sesquilinear form

$$h: X \times Y \longrightarrow K \quad \text{such that}$$

$\forall\ x, x_1, x_2 \in X, \quad y, y_1, y_2 \in Y$

and all scalars α, β following properties hold

properties

a) $h(x_1 + x_2, y) = h(x_1, y) + h(x_2, y)$

property b)

$$h(x, y_1 + y_2) = h(x, y_1) + h(x, y_2)$$

c)

c) $h(\alpha x, y) = \alpha h(x, y)$ same as inner product

d) $h(x, \beta y) = \bar{\beta}\, h(x, y)$ bilinear form

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Module No. 105

Lemma (Zero Operator)

Let X and Y be inner product spaces and $Q : X \to Y$ a bounded linear operator

Then

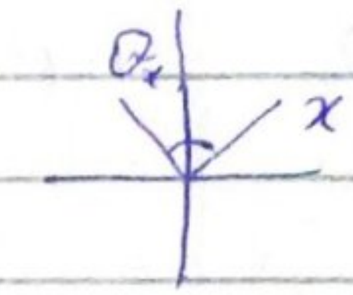
(a) $Q = 0$ if and only if $\langle Qx, y \rangle = 0$ for all $x \in X$ and $y \in Y$

(b) If $Q : X \to X$, where X is complex, and $\langle Qx, x \rangle = 0$ for all $x \in X$ then $Q = 0$

(b) is always true if X is a complex. suppose X is not complex then (b) does not hold.

Let $X = \mathbb{R}^2$

Operator Q is a rotation by angle 90°



$\langle Q_x, x \rangle = 0$

but Q is not zero. it is an rotation by an angle 90°

(a) $\Rightarrow$ $Q : X \to Y$

Take $Q = 0 \Rightarrow Qx = 0 \quad \forall \; x \in X$.

we have to show that inner product of $Qx = 0$

$\langle Qx, y \rangle = \langle 0, y \rangle = \langle 0.w, y \rangle = 0 \langle w, y \rangle = 0$

$\forall \; x \in X$

$\Leftarrow$

In reverse

Let $\langle Qx, y \rangle = 0 \quad \forall \; x, y$

$\Rightarrow Qx = 0 \; \forall x$; by the result

$\langle v, w \rangle = 0 \; \forall \; w \Rightarrow v = 0$

by def $\Rightarrow Q = 0$

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(b) $Q : X \to X$; X = complex

we need to show that if $\langle Qx, x \rangle = 0$ then $Q = 0$

Let $V \in X$;

$V = \alpha x + y$ $\quad \alpha = 1, \quad \alpha = i$

Assumption

$$\langle Qx, v \rangle = 0$$

$$\Rightarrow \quad 0 = \langle Q(\alpha x + y), \alpha x + y \rangle$$

$$= |\alpha|^2 \langle Qx, x \rangle + \langle Qy, y \rangle + \alpha \langle Qx, y \rangle + \bar{\alpha} \langle Qy, x \rangle$$

$$= \alpha \langle Qx, y \rangle + \bar{\alpha} \langle Qy, x \rangle$$

Now

$$\alpha \langle Qx, y \rangle + \bar{\alpha} \langle Qy, x \rangle = 0$$

when $\alpha = 1$ $\quad \langle Qx, y \rangle + \langle Qy, x \rangle = 0 \quad \cdots (1)$

when $\alpha = i$

$$i \langle Qx, y \rangle + \bar{i} \langle Qy, x \rangle = 0$$

$$\Rightarrow \langle Qx, y \rangle - i \langle Qy, x \rangle = 0 \quad \cdots (2)$$

adding (1) & (2)

$$\langle Qx, y \rangle = 0 \quad \text{if and only if}$$

$$\Rightarrow Q = 0$$

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Module No. 106

Theorem
(Properties of Hilbert-adjoint Operators)

Let H_1 and H_2 be Hilbert spaces

$S : H_1 \to H_2$ and

$T : H_1 \to H_2$ bounded linear operators

and α any scalar. Then we have

a) $\langle T^* y, x \rangle = \langle y, Tx \rangle$

$x \in H_1, \; y \in H_2$)

b) $(S+T)^* = S^* + T^*$

c) $(\alpha T)^* = \bar{\alpha} T^*$

d) $(T^*)^* = T$

e) $\|T^* T\| = \|T T^*\| = \|T\|^2$

f) $T^* T = 0 \iff T = 0$

g) $(ST)^* = T^* S^*$ assuming $H_2 = H_1$)

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Module No. 107

Self-adjoint, Unitary and normal ope.

Self-adjoint operator.

Def:

A bounded linear operator $T: H \to H$ on a Hilbert space H is said to be self-adjoint or Hermitian if

$$T^* = T$$

Unitary:

A bounded linear operator $T: H \to H$ on a Hilbert space H is said to be unitary if T is bijective and

$$T^* = T^{-1}$$

i) bijective. ii) $T^* = T^{-1}$

Normal Operators

A bounded linear operator $T: H \to H$ on a Hilbert space H is said to be normal if

$$TT^* = T^*T$$

Self-adjoint $T^* = T$

unitary (bijective) $T^* = T^{-1}$

Normal $TT^* = T^*T$

Hilbert-adjoint $\langle Tx, y \rangle = \langle x, T^*y \rangle$

So, if an operator is self adjoint then it is normal and

if an operator is unitary then it is normal.

Self-adjoint $\to$ Normal

Unitary $\to$ Normal

is Normal $\leftarrow$ Self-adjoint, Unitary

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A normal operator need not to be unitary

$$I : H \to H$$

$T = 2i\,I$, $T^* = -2i\,I$

$TT^* = T^*T = 4I$ normal

$T \neq T^*$

$\Rightarrow$ T is not self-adjoint

$T^{-1} = \frac{-1}{2} i\,I \neq T^*$

$\Rightarrow$ not unitary

Note: So, a normal operator is not self-adjoint and unitary.

Modul # 108
Examples

We consider $\mathbb{C}^n$ (column vectors with complex) with the inner product

$$\langle x, y \rangle = x^T \bar{y}$$

$$\langle x, y \rangle = [x_1, \ldots, x_n] \begin{bmatrix} \bar{y}_1 \\ \vdots \\ \bar{y}_n \end{bmatrix}$$

$$= [x_1 \bar{y}_1 + \cdots + x_n \bar{y}_n]$$

$$y \begin{bmatrix} y_1 \\ \vdots \\ y_2 \end{bmatrix}, y_i \in \mathbb{C} \quad \left(\begin{array}{c} x \in \mathbb{C}^n \\ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, x_i \in \mathbb{C} \end{array} \right.$$

Let $T : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ be a bounded linear operator. We can represent T and Hilbert-adjoint by T^* ---

by two matrices say A, B

$T_x = Ax$

$T_y^* = Bx$

$$\langle T_x, y \rangle = (Ax)^T \bar{y}$$

$$= x^T A^T \bar{y}$$

$$\langle x, T^* y \rangle = x^T \bar{B}\, \bar{y}$$

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as $\langle Tx, y \rangle = \langle x, T^* y \rangle$

so $x^T A^T \bar{y} = x^T \bar{B} \bar{y}$

$$A^T = \bar{B}$$

$$\Rightarrow \quad \bar{A}^T = B$$

Example

If a basis for C^n is given and a linear operator on C^n is represented by a certain matrix, then its Hilbert-adjoint operator is represented by the complex conjugate transpose of that matrix.

$$\bar{A}^T = B$$

$\bar{A}$: complex conjugate of linear operator matrix

A^T: Hilbert-adjoint operator

Consequently representing matrices are

- Hermitian if T is self-adjoint (Hermitian)
- Unitary if T is unitary.
- normal if T is normal

Similarly, for a linear operator $T: \mathbb{R}^n \to \mathbb{R}^n$, representing matrices are

Real symmetric if T is self-adjoint.

Orthogonal if T is unitary.

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Module # 109

Properties :

a) A square matrix $A = (a_{jk})$ is said to be

Hermitian if $\bar{A}^T = A$

skew-Hermitian if $\bar{A}^T = -A$

Unitary if $\bar{A}^T = A^{-1}$

normal if $A\bar{A}^T = \bar{A}^T A$

b) A real square matrix $A = (a_{jk})$ is said to be

(Real) symmetric if $A^T = A$

(Real) skew-symmetric if $A^T = -A$

Orthogonal if $A^T = A^{-1}$

Example: 1)

$$A = \begin{bmatrix} 2 & 2+i & 4 \\ 2-i & 3 & i \\ 4 & -i & 1 \end{bmatrix}, \quad \bar{A} = \begin{bmatrix} 2 & 2-i & 4 \\ 2+i & 3 & -i \\ 4 & i & 1 \end{bmatrix}$$

$$\bar{A}^T = \begin{bmatrix} 2 & 2+i & 4 \\ 2-i & 3 & i \\ 4 & -i & 1 \end{bmatrix} = A$$

So, A is Hermitian

Now unitary $\bar{A}^T = A^{-1} \Rightarrow A\bar{A}^T = I$

Example: 2)

$$A = \frac{1}{2}\begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 1-i & 1+i \\ 1+i & 1-i \end{bmatrix}, \quad \bar{A}^T = \begin{bmatrix} 1-i & 1+i \\ 1+i & 1-i \end{bmatrix}$$

$$A\bar{A}^T = \left(\frac{1}{2}\begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}\right)\left(\frac{1}{2}\begin{bmatrix} 1-i & 1+i \\ 1+i & 1-i \end{bmatrix}\right)$$

$$= \frac{1}{4}\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

So, A is unitary

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Example 3)

$$A = \begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & -4 \\ 1 & 4 & 0 \end{bmatrix}, \quad A^T = \begin{bmatrix} 0 & -2 & 1 \\ 2 & 0 & 4 \\ -1 & -4 & 0 \end{bmatrix}$$

$$-A = \begin{bmatrix} 0 & -2 & 1 \\ 2 & 0 & 4 \\ -1 & -4 & 0 \end{bmatrix}$$

As $A^T = -A \Rightarrow A$ is real-shew symmetric

Example 4)

Orthogonal property $A^T = A^{-1} \Rightarrow AA^T = I$

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad A^T = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$AA^{-1} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So, $AA^T = I$

Hence A is orthogonal

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Module # 110

Theorem (Self-adjointness)

Let $T: H \to H$ be a bounded linear operator on a Hilbert space H. Then

(a) If T is self-adjoint, $\langle Tx, x \rangle$ is real for all $x \in H$.

(b) If H is complex and $\langle Tx, x \rangle$ is real for all $x \in H$, the operator T is self-adjoint.

Proof: (a)

If T is self-adjoint

$\Rightarrow \langle Tx, x \rangle$ is real.

If T is self-adjoint

$\Rightarrow \langle \overline{Tx, x} \rangle = \langle x, Tx \rangle$

Self-adjoint

$\langle x, Tx \rangle = \langle Tx, x \rangle$

$\Rightarrow \langle \overline{Tx, x} \rangle = \langle Tx, x \rangle$

$\Rightarrow \langle Tx, x \rangle$ is real

b)

H is complex, $\langle Tx, x \rangle$ is real $\forall x$, then

$\langle Tx, x \rangle = \langle \overline{Tx, x} \rangle$

$= \langle \overline{x, T^* x} \rangle \quad \because \langle Tx, x \rangle = \langle x, T^* x \rangle$

$= \langle T^* x, x \rangle$

$\Rightarrow \langle (T - T^*)x, x \rangle = 0 \quad \because \langle Qx, y \rangle = 0$

$\Rightarrow T - T^* = 0 \quad Q = 0$

$\Rightarrow T = T^*$

We assume that H is complex and get that T is self-adjoint

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MTH # 112

Theorem (self-adjointness of product)

The product of two bounded self-adjoint linear operators S and T on a Hilbert space H is self-adjoint
if and only if the operator commute.

i.e $ST = TS$

as condition is if and only if so we prove in both direction.

i) Assume product to self-adjoint and prove commute
ii) Assume product to commute and prove self-adjoint.

Proof:

$(ST)^*$

Hilbert-adjoint operator of product (ST) $S: H \to H$, $T: H \to H$

we have seen property $(ST)^* = T^* S^*$

„⇒" Let ST is selfadjoint

$\Rightarrow (ST)^* = ST$

$T^* S^* = ST$

as $S^* = S$ and $T^* = T$

$\Rightarrow TS = ST$ commut. --①

Conversely

„⇐"

we assume that $ST = TS$ commute

$\Rightarrow (ST)^* = T^* S^*$

$= TS = ST$

$(ST)^* = ST$ --②

$\Rightarrow$ self-adjoint.

Hence from ① & ②

$ST = TS$

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Module # 112

Theorem (Sequences of self-adjoint operators)

Let (T_n) be a sequence of bounded self-adjoint linear operators $T_n : H \to H$ on a Hilbert space H.

Suppose that (T_n) converges, say,

$T_n \to T$ that is

$$\| T_n - T \| \longrightarrow 0$$

where $\|\cdot\|$ is the norm on the space $B(H,H)$. Then the limit operator T is a bounded self-adjoint linear operator on H

$B(H,H)$ is the space of all bounded linear operators define from H to H.

Proof:

We need to show $T_n \to T$,

T is self-adjoint

i-e $T^* = T$ ..① ← both are same

same as showing $\|T^* - T\| = 0$ ..②

① & ② are same.

To prove $\|T^* - T\| = 0$,

$$\|T_n^* - T^*\| = \|(T_n - T)^*\| \qquad \because (S+T)^* = S^* + T^*$$

$$= \|T_n - T\| \qquad \because \|T^*\| = \|T^*\|$$

Now

$$\|T_n^* - T^*\| = \|T_n - T\| \quad \cdots\cdots ③$$

by triangul inequality on $B(H,H)$

$$\|T - T^*\| = \|T - T_n + T_n - T_n^* + T_n^* - T^*\|$$

$$\leq \|T - T_n\| + \underbrace{\|T_n - T_n^*\|}_{0} + \|T_n^* - T^*\|$$

eq ③

we know that (T_n) seques of self-adjoint

$T_n = T_n^*$

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$$\|T - T^*\| \leq \|T - T_n\| + \|T_n^* - T_n^*\| + \|T_n - T\|$$

$$\leq \|T - T_n\| + \|T_n - T\|$$

$$= \|T_n - T\| + \|T_n - T$$

$$= 2\|T_n - T\|$$

So $\|T - T^*\| \leq 2\|T_n - T\|$

$\|T - T^*\| \leq 0$ as given $\|T_n - T\| \to 0$

$\Rightarrow \|T - T^*\| = 0$

$\Rightarrow T - T^* = 0 \Rightarrow T = T^*$

Our suppsition was $T_n - T \to 0$ $T_n \to T$

$\Rightarrow T = T^* \Rightarrow$

$\Rightarrow T$ is sel-adjoint

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Module # 113

Previously the operators was starting with self-adjoints. Now we start with unitary

Let the operators
$U: H \to H$ and
$V: H \to H$ be unitary;
here H is Hilbert space. Then

a) U is isometric, thus
a) $\|Ux\| = \|x\|$ for all $x \in H$

b) $\|U\| = 1$ provided $H \neq 0$

c) $U^{-1} (= U^*)$ is unitary

d) UV is unitary

e) U is normal

Proof: U is isometric

a)

$$\|Ux\|^2 = \langle Ux, Ux \rangle$$
$$= \langle x, U^*Ux \rangle \quad \text{using } \langle Ux, x \rangle = \langle x, U^*x \rangle$$
$$= \langle x, Ix \rangle \quad \text{as } U \text{ is unitary so } U^* = U^{-1} \text{ & } U^*U = I$$
$$= \langle x, x \rangle$$
$$= \|x\|^2$$

$$\Rightarrow \|Ux\|^2 = \|x\|^2$$

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b)

c) U^{-1} is unitary (bijective + $U^* = U^{-1}$

Proof:

since U^* is bijective

$\Rightarrow$ U^{-1} is bijective.

$(U^{-1})^* = (U^*)^*$ $\quad \because U^{**} = U$

$= U = (U^{-1})^{-1}$

so $(U^{-1})^* = (U^{-1})^{-1}$

$\Rightarrow$ U^{-1} is unitary as it bijective & $(U^{-1})^* = (U^{-1})^{-1}$

d) UV is unitary

Proof: U, V are unitary operator

We need to show two thing U, V both bijective

$U^* = U^{-1}$, $V^* = V^{-1}$

i) bijective

ii) $(UV)^* = (UV)^{-1}$

as U and V are bijective, so there product is also bijective. UV is bijective

ii) $(UV)^* = V^* U^*$ using property $(ST)^* = T^* S^*$

$= V^{-1} U^{-1}$ $\because V^* = V^{-1}$ & $U^* = U^{-1}$

$= (UV)^{-1}$

$(UV)^* = (UV)^{-1}$

Hence UV is unitar

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e To show U is normal

Proof.

$$UU^* = U^*U$$

$$U^* = U^{-1}$$

$$\Rightarrow UU^{-1} = U^{-1}U$$

$$I = I$$

Hence U is normal

f

Proof:

A bounded linear operator T on a complex Hilbert space H is unitary if and only if

i) T is isometric

ii) Surjective

" $\Rightarrow$ "

Suppose T is isometric and surjective to show T is unitary.

Isometric means

$\Rightarrow$ T is injective

From our supposition (T is surjective) and from isometric (T is injective) we can write that T is bijective.

Now to show tha $T^* = T^{-1}$

Again by isometry

$$\langle T^*Tx, x\rangle = \langle Tx, Tx\rangle$$

$$= \langle x, x\rangle = \langle Ix, x\rangle$$

$$\langle T^*Tx, x\rangle - \langle Ix, x\rangle = 0$$

$$\langle (T^*T, I)x, x\rangle = 0$$

Since it is complex, using previous result

$$T^*T - I = 0$$

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$T^*T = I$ — ①

Now $TT^* = TT^*I$

$= TT^*(TT^{-1})$

$= T(T^*T)T^{-1}$ using $T^*T = I$

$= T(I)T^{-1}$

$= TT^{-1}$

$= I$ — ②

$\Rightarrow TT^* = I$

From ① and ②

$TT^* = T^*T = I$

"$\Leftarrow$" **Conversely**,

Suppose T is unitary

we have to show T is surjective + isometry.

As T is unitary it means

i) T is bijective } by supposition.
ii) $T^* = T^{-1}$ }

iii) From part a) T is isometry.

iv)

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